

Fast and Robust Phase Equilibrium Computations for $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ Mixtures Using the Electrolyte Cubic-Plus-Association Equation of State

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Abstract

We present phase stability and flash calculations for $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ mixtures using the electrolyte Cubic-Plus-Association (eCPA) equation of state. In this ternary system, H_2O partitioning into the CO_2 -rich phase concentrates NaCl in the aqueous phase, raising the equilibrium molality m_s^{aq} above the feed value and suppressing CO_2 solubility. Modeling this feed-dependent salting-out effect requires full stability+flash rather than a fixed VLE lookup. Starting from the eCPA thermodynamic framework, we develop: (i) a generalized salt-free CPA flash routine with Michelsen tangent-plane distance (TPD) stability tests using six initial guesses and accelerated successive-substitution iterations (SSI); (ii) an extension to the full ternary system via analytical elimination of m_s^{aq} within each SSI step; (iii) a pre-computed three-dimensional solution table spanning $T = 0\text{--}360^\circ\text{C}$, $P = 1\text{--}2000\text{ bar}$, and $m_s = 0\text{--}6\text{ mol kg}^{-1}$ that provides warm-start initial guesses reducing flash cost by $3.2\times$ for the ternary system and more than two orders of magnitude for the salt-free CPA binary (cold-start six-trial robust algorithm versus warm-started SSI); and (iv) a simplified flash exploiting $y_{\text{H}_2\text{O}} \approx 0$, valid for $T \lesssim 80^\circ\text{C}$ with errors below 2% in dissolved CO_2 molality. Accelerated SSI reduces the mean flash iteration count by 46% (25.4 to 13.7) versus standard SSI; analytical Newton polish after warm-started SSI further reduces the mean to 6.9 iterations ($\sim 20\%$ wall-time saving). Analyt-

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ical Jacobian-based Newton–Raphson inner solvers reduce the per-iteration cost of CPA/eCPA to only $\sim 3\times$ that of a plain cubic EoS, and are 6–13 $\times$ faster than cold-start alternatives. The six-guess stability test detects all two-phase conditions without false negatives. Validation against >1100 experimental data points achieves 100% flash convergence with average absolute relative errors (AARE) of 9.6% for CO₂ solubility in pure water (6.5% under subsurface storage conditions) and 24.4% for H₂O content in the CO₂-rich phase (17.4% after excluding flagged outlier datasets); 6.6% for CO₂ molality in NaCl brine and 6.2% across all converged points ($T = 5\text{--}350\text{ }^{\circ}\text{C}$); and 0.30% for pure-water density over 475 IAPWS-95 reference conditions ($T = 0\text{--}350\text{ }^{\circ}\text{C}$, $P = 1\text{--}2\,000\text{ bar}$) using a temperature-dependent P  neloux volume shift for H₂O (0.12% under typical subsurface storage conditions).

Keywords: CO₂ sequestration, eCPA equation of state, phase stability, flash calculation, accelerated successive substitution, reservoir simulation

1. Introduction

Geological storage of CO₂ in saline aquifers and depleted hydrocarbon reservoirs is one of the most important near-term options for large-scale carbon capture and storage [1, 2]. Geothermal energy extraction presents an equally compelling application [3]: enhanced geothermal systems and hydrothermal reservoirs operate at temperatures often above 227 $^{\circ}\text{C}$, where CO₂ can serve as a working fluid or co-migrate with brine, and where the phase behavior of the CO₂ + H₂O + NaCl system must be accurately described to model fluid circulation, energy extraction efficiency, and mineral scaling. In both applications, accurate modeling of CO₂+brine phase behavior is a prerequisite for reliable reservoir simulation: the equilibrium compositions of the CO₂-rich and aqueous phases govern injectivity, trapping efficiency, and caprock integrity, while accurate density predictions are needed for fluid flow and pressure calculations. The high-temperature range ($T > 227\text{ }^{\circ}\text{C}$) is particularly relevant to geothermal applications, motivating the wide temperature coverage of the present model.

The equilibrium of CO₂ with brine is challenging to model. It spans a wide range of conditions—temperatures from near-ambient ($\sim 10\text{ }^{\circ}\text{C}$) to supercritical and hydrothermal ($> 327\text{ }^{\circ}\text{C}$), pressures from hydrostatic (1 bar) to lithostatic ($> 1000\text{ bar}$), and salinities from fresh water to near-saturation ($> 6\text{ mol kg}^{-1}\text{ NaCl}$)—and involves both association interactions (hydrogen

22 bonding in water), CO₂–water cross-association, and electrostatic effects of
23 dissolved ions.

24 Cubic-Plus-Association (CPA) equations of state [4] successfully combine
25 a cubic (Soave–Redlich–Kwong, SRK) repulsive / dispersion term with a
26 Wertheim association term [5] and have been widely applied to systems with
27 hydrogen-bonding fluids. Considering the *cross*-association between CO₂ and
28 H₂O molecules further improved the predictive powers of CPA considerably
29 [6]. The electrolyte CPA (eCPA) extension [7] adds Debye–Hückel and Born
30 electrostatic terms to describe ion–solvent and ion–ion interactions. Kontogeorgis et al. [8] provide a review of different versions of the eCPA EOS.

32 Coelho et al. [9] recently parameterized eCPA for the CO₂ + H₂O + NaCl
33 ternary system with temperature-dependent binary interaction parameters,
34 achieving a 74% reduction in CO₂ solubility error in brine compared to the
35 prior model of Maribo-Mogensen et al. [7], which neglected cross-association
36 (AARE: 6.9% vs. 26.6% over 660 data points from 10 to 450 °C).

37 Despite the availability of well-parameterized thermodynamic models,
38 published eCPA work has largely focused on the *equilibrium conditions* —
39 equality of fugacities across phases — without solving the accompanying ma-
40 terial balance equations. For the salt-free CO₂ + H₂O binary this omission
41 is less consequential: phase compositions are fixed by (T, P) alone (Gibbs
42 phase rule), so material balance reduces to a trivial scalar equation for
43 the phase fraction once equilibrium is established. For the salt-containing
44 ternary, however, the equilibrium aqueous molality m_s^{aq} itself depends on the
45 phase fraction through the salting-out effect, coupling material balance and
46 equilibrium into a nonlinear system that cannot be decoupled without in-
47 troducing thermodynamic inconsistency. Even for the simpler CPA binary,
48 phase-split algorithms benefit from more efficient root-finding strategies and
49 robust initialization than have been reported. To our knowledge, no exist-
50 ing work applies phase-split calculations to eCPA with the algorithmic rigor
51 — robust tangent-plane stability testing, consistent material balance, and
52 efficient iteration — routinely demanded in hydrocarbon flash computations.

53 A key thermodynamic subtlety compounds this challenge. For a binary
54 mixture such as CO₂ + H₂O, the Gibbs phase rule fixes the compositions of
55 the coexisting phases at any given T and P : only the phase *amounts* depend
56 on the overall CO₂ feed mole fraction, so a simple vapor–liquid equilibrium
57 (VLE) table indexed on (T, P) fully characterizes the system. In the ternary
58 CO₂ + H₂O + NaCl system this no longer holds. As CO₂ partitions into
59 the CO₂-rich phase it carries along some water, so the equilibrium aqueous

NaCl molality m_s^{aq} exceeds the feed molality m_s by an amount that grows with the CO_2 phase fraction β —the well-known salting-out effect. Because m_s^{aq} depends on the overall compositions, $\mathbf{z}$, through the phase split, the equilibrium phase compositions themselves depend on $\mathbf{z}$: a VLE table at fixed feed salinity is thermodynamically inconsistent for the ternary system, and full stability+flash is required to resolve the self-consistent $(\beta, m_s^{\text{aq}}, \mathbf{x}, \mathbf{y})$ output for every $(T, P, \mathbf{z}, m_s)$ query. Reservoir simulators therefore need flash calculations that are both *robust*—returning the correct phase split across the entire physically relevant parameter space—and *fast*—executing in milliseconds per grid cell to be compatible with time-stepping over millions of cells. Achieving this for an EoS as complex as eCPA (which requires iterative solution of both the cubic compressibility equation and the nonlinear association equations) is non-trivial.

The principal objective of this work is to develop an efficient, robust, and thermodynamically consistent phase-split algorithm for CO_2 +brine systems based on eCPA, bringing the algorithmic standard of practice from hydrocarbon flash computations to this more complex, salt-containing electrolyte system. The specific contributions are:

1. A material-balance formulation for the salt-containing aqueous phase and its numerical implementation: the equilibrium aqueous molality m_s^{aq} is determined self-consistently within the SSI loop, replacing an earlier robust but slow outer root-finding loop over molality and eliminating the thermodynamic inconsistency of fixing m_s^{aq} at the feed value.
2. Analytical Jacobian-based Newton–Raphson solvers for both the CPA and eCPA equations of state that reduce the inner-loop cost to only $\sim 3\times$ that of a plain cubic EoS with an analytical Z -root solution, rather than the one-to-two orders of magnitude typically assumed.
3. A robust, accelerated successive-substitution (SSI) flash for the $\text{CO}_2 + \text{H}_2\text{O}$ binary with Michelsen tangent-plane distance (TPD) stability testing [10] using six initial guesses [11] and dominant-eigenvalue step-size control [11], validated across 29,412 (T, P, z) conditions with 100% convergence and a $\sim 2\times$ iteration-count reduction over standard SSI.
4. A precomputed three-dimensional (T, P, m_s) solution table with 505,400 grid points covering $T = 0\text{--}360^\circ\text{C}$, $P = 1\text{--}2000$ bar, and $m_s = 0\text{--}6\text{ mol kg}^{-1}$, providing warm-start initial guesses that reduce iteration counts from ~ 12 to ~ 3 ($3.2\times$ wall-time speedup) and are suitable for integration into reservoir simulators.

5. A complete open implementation of the full eCPA equation of state of Coelho et al. [9], including Debye–Hückel, Born, association, and permittivity terms, validated against the original parameterization over $T = 10\text{--}455^\circ\text{C}$.
6. Several algorithmic optimizations: multi-core parallelization for solution-table construction, phase-envelope tracing, and stability mapping; temperature and salinity continuation within the table build to maximize convergence rates; and a logarithmic pressure axis for the interpolation grid.
7. Systematic validation against experimental vapor–liquid equilibrium (VLE) and density data, demonstrating 0.30% AARE for pure-water density over 475 IAPWS-95 reference conditions ($T = 0\text{--}350^\circ\text{C}$, $P = 1\text{--}2000\text{ bar}$) using a temperature-dependent Péneloux volume shift for H_2O .
8. A minimal but complete IMPEC compositional reservoir simulator (both Mixed Finite Element and Two-Point-Flux-Approximation options, two-phase Darcy flow, Corey relative permeabilities) that integrates the fast CPA and eCPA flash routines and demonstrates 8,840–14,270 flash calls s^{-1} on a 2500-cell grid (80–87% of step cost), with a discussion of how the routine can be embedded in existing simulators such as MRST [12], OPM Flow [13], and DuMu x [14].

The remainder of this paper is organized as follows. Section 2 presents the CPA and eCPA thermodynamic framework. Section 3 describes the phase stability test and flash algorithms, including the hierarchical multi-guess strategy, the accelerated SSI, the precomputed solution table, and a simplified flash exploiting $y_{\text{H}_2\text{O}} \approx 0$. Section 4 validates the implementation against experimental $\text{CO}_2 + \text{H}_2\text{O}$ and $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ VLE and density data. Section 5 analyzes computational performance of the inner and outer Newton–Raphson solvers and the warm-start speedup. Section 6 demonstrates the flash routines within a minimal IMPEC reservoir simulator. Conclusions are given in Section 7. Additional technical details and supporting figures are provided in Supplemental Information.

2. Thermodynamic Framework

Throughout this work the two volatile components are H_2O and CO_2 ; Na^+ and Cl^- enter only through the molality m_s and the ionic parameters

of the electrostatic terms. NaCl is treated as a single molecular species; extension to multiple-salt brines on an ionic basis is discussed in Section 7.

Mole fractions in the aqueous phase are denoted x_i and in the CO₂-rich phase y_i , with equilibrium ratios (K-values) $K_i = y_i/x_i$. Component subscripts use the full species name (e.g. $x_{\text{H}_2\text{O}}$, y_{CO_2} , $K_{\text{H}_2\text{O}}$, z_{CO_2}); phase superscripts use ^{aq} for the aqueous phase and ^c for the CO₂-rich phase (e.g. Z^{aq} , $\hat{\phi}_i^{\text{aq}}$, $\hat{\phi}_i^{\text{c}}$). m_s with subscript s denotes NaCl molality. A complete list of symbols is given in the Nomenclature section of the Supporting Information.

2.1. CPA equation of state

For the salt-free CO₂ + H₂O binary the residual Helmholtz energy per mole is

$$A^r = A^{\text{SRK}} + A^{\text{assoc}}, \quad (1)$$

where A^{SRK} is the Soave–Redlich–Kwong [15] physical term and A^{assoc} is the Wertheim association term [5].

The SRK compressibility factor satisfies

$$Z = \frac{v}{v-b} - \frac{a(T)}{RT} \frac{1}{v+b}, \quad (2)$$

where v is the molar volume and x_i is the mole fraction of species i ; $a(T) = \sum_i \sum_j x_i x_j (1 - k_{ij}) \sqrt{a_i a_j}$ and $b = \sum_i x_i b_i$ are the mixture attractive and co-volume parameters, respectively, with sums over molecular species $i, j \in \{\text{H}_2\text{O}, \text{CO}_2\}$ (and additionally Na^+ , Cl^- for brines). The Soave temperature dependence is $a_i(T) = a_{0i} [1 + c_{1i} (1 - \sqrt{T/T_{c,i}})]^2$. The Peng–Robinson (PR) equation of state [16] can serve equally well as the cubic backbone; it has been used with cross-association in the CPA framework for water-containing mixtures [6, 17, 18, 19, 20]. We follow Coelho et al. [9] in using SRK throughout the present work.

The association term accounts for hydrogen bonding in water via the 4C scheme (two proton donors and two proton acceptors), and includes cross-association between CO₂ and H₂O [6]. The sum runs over molecular species only ($i \in \{\text{H}_2\text{O}, \text{CO}_2\}$); ions carry no association sites:

$$\frac{A^{\text{assoc}}}{NRT} = \sum_i x_i \sum_{A \in i} \left(\ln \chi^{A_i} - \frac{\chi^{A_i}}{2} + \frac{1}{2} \right), \quad (3)$$

where χ^{A_i} is the fraction of molecules i not bonded at site A :

$$\chi^{A_i} = \frac{1}{1 + c \sum_j x_j \sum_{B \in j} \chi^{B_j} \delta^{A_i B_j}}, \quad (4)$$

with $c = N_A/v$ the molecular number density (N_A is Avogadro's number) and $\delta^{A_i B_j}$ the association bonding volume between site A on species i and site B on species j . Because all sites within a given species are equivalent — H₂O has four identical sites (4C scheme) and CO₂ has one cross-association site — both the bonding fractions and the bonding volumes are the same for every site on a given species. The fractions simplify to $\chi_i \equiv \chi^{A_i} = \chi^{B_i}$ for $i \in \{\text{H}_2\text{O}, \text{CO}_2\}$, and the bonding volume reduces to a single scalar per species pair, $\delta_{ij} \equiv \delta^{A_i B_j}$. We use this simplified notation — $\chi_{\text{H}_2\text{O}}$, χ_{CO_2} , $\delta_{\text{H}_2\text{O}}$, and $\delta_{\text{CO}_2, \text{H}_2\text{O}}$ — throughout the inner Newton equations.

The CO₂–H₂O cross-association strength is proportional to water self-association [9]:

$$\delta_{\text{CO}_2, \text{H}_2\text{O}} = S_{\text{CO}_2, \text{H}_2\text{O}}(T) \delta_{\text{H}_2\text{O}}, \quad (5)$$

where $S_{\text{CO}_2, \text{H}_2\text{O}}(T)$ and the binary interaction parameter $k_{\text{CO}_2, \text{H}_2\text{O}}(T)$ both follow quadratic polynomials in $T/T_{c, \text{CO}_2}$ ($T_{c, \text{CO}_2} = 31.3^\circ\text{C}$, the CO₂ critical temperature), with coefficients from Coelho et al. [9] (Table S3 of the Supplemental Information).

The temperature dependence of both $k_{\text{CO}_2, \text{H}_2\text{O}}$ and $S_{\text{CO}_2, \text{H}_2\text{O}}$ is essential because the CO₂+H₂O system spans qualitatively different regimes across the validated temperature range. Below $T_{c, \text{CO}_2} = 31.3^\circ\text{C}$, CO₂ can be subcritical liquid and its fugacity is governed primarily by vapor-pressure equilibrium. Between $\sim 31^\circ\text{C}$ and $\sim 177^\circ\text{C}$, CO₂ is supercritical and the aqueous solubility decreases steeply with temperature. Above $\sim 227^\circ\text{C}$, the CO₂-rich phase acquires substantial water content (up to 20–25 mol% at $T = 227^\circ\text{C}$ and $P = 200$ bar), making accurate modeling of both phases equally important. A single set of temperature-independent binary interaction parameters cannot simultaneously describe all three regimes; the quadratic polynomial form of Coelho et al. [9] provides sufficient flexibility while remaining analytically tractable.

2.2. *eCPA equation of state*

For the ternary system, two electrostatic terms are added [7]:

$$A^r = A^{\text{SRK}} + A^{\text{assoc}} + A^{\text{DH}} + A^{\text{Born}}, \quad (6)$$

where A^{DH} and A^{Born} are the Debye–Hückel and Born solvation contributions, respectively. The presence of NaCl (Na⁺ + Cl[−] ion pair) modifies the dielectric constant ϵ_r of the aqueous phase, which enters both electrostatic

192 terms. The permittivity is computed self-consistently from the Onsager–
 193 Kirkwood equation, linking it to the Wertheim association fractions and the
 194 molecular dipole moments [9].

195 Ion–molecule interactions in the SRK mixing rule are modeled via en-
 196 ergy parameters ΔU_{is} of the Non-Random Two-Liquid (NRTL) type with a
 197 temperature dependence of the form

$$\Delta U_{is} = \Delta U_{is}^{\text{ref}} + \alpha_{is} R \left[\left(1 - \frac{T}{T_{is}^{\alpha}} \right)^2 - \left(1 - \frac{T_{\text{ref}}}{T_{is}^{\alpha}} \right)^2 \right], \quad (7)$$

198 where $\Delta U_{is}^{\text{ref}}$ is a reference interaction energy, α_{is} is the NRTL non-randomness
 199 parameter, T_{is}^{α} is a characteristic temperature, and $T_{\text{ref}} = 25^\circ\text{C}$; all param-
 200 eters are from Coelho et al. [9]. All EoS parameters are listed in Tables S2
 201 and S3 of the Supplemental Information.

202 2.3. Density and P  neloux volume correction

203 The molar volume in each phase is $v = ZRT/P$. Cubic equations of
 204 state systematically over- or underpredict liquid densities. The P  neloux
 205 volume translation [21] corrects this without affecting phase compositions (it
 206 is isofugacity-preserving):

$$v^{\text{corr}} = v^{\text{EoS}} + \sum_i x_i c_i, \quad (8)$$

207 with the corrected density $\rho = M_{\text{mix}}/v^{\text{corr}}$. Coelho et al. [9] apply a shift
 208 $c_s = -53.5 \text{ cm}^3 \text{ mol}^{-1}$ to the NaCl species to correct brine density but that
 209 work did not consider a shift for H₂O. In this work, we also fit a temperature-
 210 dependent P  neloux volume shift for H₂O to IAPWS-95 pure-water data
 211 (Section 4.4): $c_{\text{H}_2\text{O}}(T) = \sum_k a_k T_R^k$ (degree 4, $T_R = T/T_{c,\text{H}_2\text{O}}$) with coefficients
 212 listed in Table S4 of the Supplemental Information (Section S2). No volume
 213 shift is applied to CO₂.

214 2.4. Fugacity coefficients and phase equilibrium

215 The fugacity coefficient of species i in a mixture is obtained from the
 216 residual Helmholtz energy via the standard thermodynamic identity

$$\ln \hat{\phi}_i = \left. \frac{\partial(nA^r/RT)}{\partial n_i} \right|_{T,V,n_{j \neq i}} - \ln Z. \quad (9)$$

217 Two-phase VLE equilibrium requires equal fugacities for the volatile compo-
 218 nents in both phases:

$$x_i \hat{\phi}_i^{\text{aq}}(T, P, \mathbf{x}, m_s^{\text{aq}}) = y_i \hat{\phi}_i^{\text{c}}(T, P, \mathbf{y}), \quad i \in \{\text{H}_2\text{O}, \text{CO}_2\}. \quad (10)$$

219 NaCl is non-volatile and remains entirely in the aqueous phase.

220 Evaluating $\hat{\phi}_i$ requires the compressibility Z and, for associating compo-
 221 nents, the bonding fractions χ . For the CPA binary, differentiating $A^r =$
 222 $A^{\text{SRK}} + A^{\text{assoc}}$ (Eq. (1)) with respect to n_i at constant T, V leads to a 3×3
 223 Newton system for $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ (Section 3.3). For the eCPA aqueous
 224 phase, the Debye–Hückel and Born contributions couple Z^{aq} to the relative
 225 permittivity ε_r and the association fraction $\chi_{\text{H}_2\text{O}}$, requiring a 3×3 Newton
 226 solve for $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ (Eqs. (24)–(26), Section 3.4).

227 When m_s^{aq} is treated as an outer unknown (Section 3.1), the H₂O material-
 228 balance residual to be zeroed is

$$R(m_s^{\text{aq}}) = m_s^{\text{aq}} - \frac{m_s z_{\text{H}_2\text{O}}}{[1 - \beta] x_{\text{H}_2\text{O}}} = 0, \quad (11)$$

229 where β and $x_{\text{H}_2\text{O}}$ are obtained by solving Eq. (10) at the trial m_s^{aq} .

230 3. Phase Stability and Flash Computations

231 *Problem statement.* Given a feed composition $\mathbf{z} = (z_{\text{H}_2\text{O}}, z_{\text{CO}_2})$ (overall mole
 232 fractions of H₂O and CO₂), temperature T , pressure P , and feed NaCl mo-
 233 lality m_s , the computational objective is to determine: (i) the phase state
 234 (single- or two-phase), (ii) the phase fraction β (moles of CO₂-rich phase
 235 per mole of feed), and (iii) the equilibrium phase compositions $\mathbf{x}$ and $\mathbf{y}$ to-
 236 gether with the self-consistent aqueous molality $m_s^{\text{aq}} \geq m_s$. Throughout this
 237 work conditions are restricted to the fluid region; salt precipitation is not
 238 considered.

239 3.1. Initial implementation: outer Brent loop over aqueous molality

240 As a first, robust implementation of the eCPA flash, we employ the
 241 Brent method [22] — a bracketed root-finding algorithm combining bisection
 242 with inverse quadratic interpolation — to find the roots of the H₂O
 243 material-balance residual $R(m_s^{\text{aq}})$ (Eq. (11)). At each trial m_s^{aq} , the isofug-
 244 acity conditions (Eq. (10)) are solved for both volatile components by an
 245 inner VLE solver, which requires evaluating the eCPA fugacity coefficients

(Eq. (9)) and hence the aqueous-phase compressibility and association state ($Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}}$) via the Newton system of Eqs. (24)–(26). Initial guesses for the inner solver are obtained from a molality-continuation cache built by marching from $m_s^{\text{aq}} = 0$ to an upper bound in small steps, requiring no initial guess for the full phase-state vector and making the approach highly robust. However, each evaluation of $R(m_s^{\text{aq}})$ requires nested Newton solves for $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$, making the outer loop expensive and ill-suited to the large number of evaluations required in a parameter-space scan or a production reservoir simulator. This motivates the development of the direct SSI approach described below.

3.2. Michelsen tangent-plane distance stability test

We implement the Michelsen tangent-plane distance (TPD) criterion [10] to classify a given $(T, P, \mathbf{z})$ as stable (single phase) or unstable (two-phase); NaCl is not an independent flash variable but enters through the molality-dependent fugacity coefficients (Eq. 9). Following Michelsen [10], we introduce the vector of *unnormalized* trial mole numbers $\mathbf{W} = (W_{\text{H}_2\text{O}}, W_{\text{CO}_2})$, which do not in general sum to unity. The normalized trial mole fractions are $w_i = W_i / \sum_j W_j$, and the TPD function is

$$\text{TPD}(\mathbf{W}) = \sum_i W_i \left[\ln W_i + \ln \hat{\phi}_i(\mathbf{w}) - d_i \right], \quad (12)$$

where the sum is over $i \in \{\text{H}_2\text{O}, \text{CO}_2\}$, $d_i = \ln z_i + \ln \hat{\phi}_i(\mathbf{z})$ are reference chemical-potential offsets evaluated at the feed composition, and $\hat{\phi}_i$ is the fugacity coefficient. At any stationary point of TPD, one can show that $\text{TPD} = 1 - \sum_i W_i$, so a negative minimum (equivalently, $\sum_i W_i > 1$ at convergence of the SSI) indicates thermodynamic instability. Initial trial compositions are specified as normalized mole fractions derived from each K-value set (e.g. for the Wilson CO_2 -rich trial, $w_{\text{H}_2\text{O}}^{(0)} = K_{\text{H}_2\text{O}} z_{\text{H}_2\text{O}} / \sum_i K_i z_i$; for the fixed trials, $y_{\text{H}_2\text{O}} = 0.01$ or $x_{\text{H}_2\text{O}} \approx 0.999$).

Following Jex et al. [11], we initialize the stability SSI from six trial K-value sets to ensure that no unstable stationary point is missed:

1. Wilson’s correlation: $K_i^W = (P_{c,i}/P) \exp[5.373(1 + \omega_i)(1 - T_{c,i}/T)]$,
2. Inverse Wilson: $K_i = 1/K_i^W$,
3. Cube-root Wilson: $K_i = (K_i^W)^{1/3}$,
4. Inverse cube-root: $K_i = (1/K_i^W)^{1/3}$,

- 278 5. CO₂-rich: $K_{\text{CO}_2} = 0.95/z_{\text{CO}_2}$, $K_{\text{H}_2\text{O}} = 0.05/z_{\text{H}_2\text{O}}$,
 279 6. H₂O-rich: $K_{\text{H}_2\text{O}} = (1 - 10^{-15})/z_{\text{H}_2\text{O}}$, $K_{\text{CO}_2} = 10^{-15}/z_{\text{CO}_2}$.

280 Each trial is iterated to convergence or a maximum of 200 SSI steps using
 281 the accelerated scheme described in Section 3.3. The mixture is classified as
 282 unstable if any trial converges to $\text{TPD} < -10^{-7}$; the K-values from the most
 283 negative TPD are passed to the flash as initial guess (Section 3.3).

284 As demonstrated in the parameter-space scan of Section 5.1, all six initial
 285 guesses play a material role: no single guess is best across the full $(T, P, \mathbf{z})$
 286 space, and all six are needed to ensure the optimal K-value initialization for
 287 the subsequent flash (Fig. 6b).

288 In practice, the stability test is the most computationally expensive step
 289 (~ 30 ms per call for the eCPA system), because it requires iterating the
 290 inner equations at each SSI step for each of the six trials. In the fast-flash
 291 algorithm described in Section 3.6, the stability test is bypassed for conditions
 292 pre-classified as two-phase by the solution table, reducing this overhead to
 293 zero for the vast majority of reservoir simulation cells.

294 *EoS root consistency in the stability reference.* The reference $d_i = \ln z_i +$
 295 $\ln \hat{\phi}_i(\mathbf{z})$ and all trial-phase evaluations use the same EoS, but the highly
 296 nonlinear CPA and eCPA equations of state can admit multiple real com-
 297 pressibility roots at a given condition, and the number of roots cannot be
 298 determined a priori. If the reference solver converges to a different root than
 299 the trial-phase solver, the resulting TPD comparison is between fugacities be-
 300 longing to physically distinct solution branches, and the sign of TPD carries
 301 no thermodynamic meaning. This root-mixing artefact is most pronounced
 302 at pressures below the saturation pressure of water, $P < P_{\text{sat}}(\text{H}_2\text{O}, T)$: an
 303 H₂O-rich feed composition (large $z_{\text{H}_2\text{O}}$) can simultaneously support a liquid-
 304 like root ($Z \ll 1$) and a gas-like root ($Z \approx 1$), among others. A naive
 305 implementation that returns the *first* valid root found risks root-mixing: for
 306 H₂O-rich compositions at low pressure the reference may converge to the
 307 liquid root ($Z \ll 1$), while CO₂-rich trial iterates at small $y_{\text{H}_2\text{O}}$ converge
 308 to the gas root ($Z \approx 1$), producing spuriously large negative TPD values
 309 and false two-phase predictions. Our implementation avoids this by collect-
 310 ing *all* distinct valid roots from multiple cold starts and selecting the one
 311 that minimizes the residual Gibbs energy $G^{\text{res}}/RT = \sum_i x_i \ln \hat{\phi}_i$. This is
 312 a purely thermodynamic criterion: the minimum- G root is the thermody-
 313 namically stable single-phase state at any $(T, P, \mathbf{x})$, independent of initial

guess ordering or any pressure threshold. In the parameter-space scan (Section 5.1), this criterion eliminates nine false-positive two-phase predictions at $T = 190\text{--}220^\circ\text{C}$, $P = 10$ bar (below P_{sat} for those temperatures), conditions at which no genuine two-phase equilibrium exists according to the EoS.

3.3. Successive-substitution isothermal flash

For two-phase conditions, the isofugacity conditions (Eq. (10)) are iterated by successive substitution on the K-values $K_i = y_i/x_i$. At each iteration k , the fugacity-ratio residual vector is

$$g_i^{(k)} = \ln \hat{\phi}_i^c(y^{(k)}) + \ln y_i^{(k)} - \ln \hat{\phi}_i^{\text{aq}}(x^{(k)}) - \ln x_i^{(k)}, \quad (13)$$

and the K-value update is

$$\ln K_i^{(k+1)} = \ln K_i^{(k)} - m^{(k)} g_i^{(k)}, \quad (14)$$

where $m^{(k)}$ is a step-size acceleration factor, discussed next.

Accelerated SSI. Following the dominant-eigenvalue acceleration of Jex et al. [11], the step size is

$$m^{(k)} = \left| \frac{\mathbf{g}^{(k-1)\top} \mathbf{g}^{(k-1)}}{\mathbf{g}^{(k-1)\top} [\mathbf{g}^{(k-1)} - \mathbf{g}^{(k)}]} \cdot m^{(k-1)} \right|, \quad m^{(k)} \in [1, 1/L], \quad (15)$$

with $L = 0.1$ (so $m^{(k)} \leq 10$) for two-phase systems, and $m^{(1)} = 1$ (standard SSI on the first step). The formula estimates the dominant eigenvalue of the iteration map and extrapolates the K-value update along the slowest-decaying direction, dramatically reducing the number of iterations needed for convergence. Following Jex et al. [11], we use the identity matrix as the scaling operator $\mathbf{U}^{-1}$ in the inner product, which is exact for two-component systems. The accelerated step $m^{(k)} g_i^{(k)}$ is clipped to $[-5, +5]$ in log-K space to prevent overshooting.

Convergence is declared when $\|\mathbf{g}^{(k)}\| < \tau$ with tolerance $\tau = 10^{-10}$.

Newton polish. Once the SSI residual falls below a switchover threshold τ_{switch} (default 10^{-4}), the solver switches to a Newton–Raphson step for quadratic convergence. At each Newton step, we solve the 2×2 linear system

$$\mathbf{J}^{(k)} \Delta \ln \mathbf{K}^{(k)} = -\mathbf{g}^{(k)}, \quad (16)$$

where the Jacobian $\mathbf{J} = \partial \mathbf{g} / \partial \ln \mathbf{K}$ is computed analytically. Each column of $\mathbf{J}$ requires the composition derivative $\partial \ln \hat{\phi}_i / \partial x_j$, which is obtained via the implicit function theorem applied to the three coupled inner equations for $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$: a 3×3 linear system is solved analytically at the converged SSI state, yielding dZ/dx , $d\chi_{\text{H}_2\text{O}}/dx$, and $d\chi_{\text{CO}_2}/dx$, from which $\partial \ln \hat{\phi}_i / \partial x_j$ follows by the chain rule. The Newton step is clipped to $[-5, +5]$ in $\ln K$ space. If Newton fails to reduce $\|\mathbf{g}\|$ or if the Jacobian is singular, the solver reverts to the pre-Newton state and continues with accelerated SSI, ensuring robustness. Across 9,825 two-phase conditions, Newton polish is triggered at 85.6% of points and reduces the mean iteration count from 9.8 (SSI only) to 6.9 (4.8 SSI + 2.1 Newton); see Table 4 and Fig. S10.

The CO_2 -rich phase mole fraction β (the fraction of total moles in the CO_2 -rich phase) is obtained at each step by solving the Rachford–Rice equation:

$$\sum_i \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0, \quad (17)$$

where the sum is over $i \in \{\text{H}_2\text{O}, \text{CO}_2\}$ and z_i are the overall feed mole fractions. Initial K-values are taken from Wilson’s correlation, from the stability test (Section 3.2), or from the solution table (Section 3.6.1).

Hierarchical flash algorithm. The stability test and flash are combined in a hierarchical algorithm [11]: (1) run the TPD stability test with all six initial guesses; if stable, return single-phase; (2) use K-values from the most negative TPD trial as the initial guess for the accelerated SSI flash; (3) if the flash fails, fall back to Wilson K initialization. This strategy provides both robustness (100% convergence across all 9,825 two-phase conditions in the parameter-space scan, Section 5.1) and efficiency (stability-informed K-values reduce flash iterations by an additional 8% compared to Wilson initialization).

For cold-start SSI (without a table guess), the full unit step in Eq. (14) ($m^{(k)} = 1$) can overshoot near difficult conditions such as the CO_2 critical region, causing oscillation or slow divergence. The non-accelerated fallback path therefore applies *under-relaxation*: each K-value update is multiplied by a damping factor $\omega = 0.7$,

$$\ln K_i^{(k+1)} = \ln K_i^{(k)} - \omega g_i^{(k)},$$

reducing each step to 70% of the standard correction. This is conceptually the opposite of the Jex extrapolation factor $m^{(k)} \geq 1$ (Eq. (15)), which *amplifies* the update to accelerate convergence when the iteration is already

371 on a smooth trajectory toward the solution; the damping factor $\omega < 1$ in-
 372 stead *stabilizes* the iteration when the starting point is far from the solution.
 373 Warm-started SSI (from a table guess) begins already close to the converged
 374 state and converges reliably with the full step ($\omega = 1.0$) in 3–4 iterations.

375 *Inner Newton–Raphson solve for Z and χ (CPA).* Within each SSI iteration,
 376 the CPA fugacity coefficients require simultaneous solution of the compress-
 377 ibility factor Z and the association site fractions χ for each phase. These
 378 quantities are mutually dependent: Z sets the molar volume and hence the
 379 packing fraction η , which enters the radial distribution function $g(\eta)$ and
 380 therefore the association strength Δ ; the association fractions χ in turn feed
 381 back into the compressibility equation through the Wertheim term. For the
 382 CO₂–H₂O binary, the inner problem comprises three coupled equations per
 383 phase (written here for the aqueous phase with mole fractions x_i ; the CO₂-
 384 rich phase uses the identical system with y_i in place of x_i):

$$F_1 = Z - \frac{Z}{Z - B} + \frac{A}{Z + B} - \left(1 + \frac{\eta}{g} \frac{dg}{d\eta}\right) [x_{\text{H}_2\text{O}}(\chi_{\text{H}_2\text{O}} - 1) + x_{\text{CO}_2}(\chi_{\text{CO}_2} - 1)] = 0, \quad (18)$$

$$F_2 = c_3 \chi_{\text{H}_2\text{O}}^3 + c_2 \chi_{\text{H}_2\text{O}}^2 + c_1 \chi_{\text{H}_2\text{O}} + c_0 = 0, \quad (19)$$

$$F_3 = \chi_{\text{CO}_2} (Z + 2x_{\text{H}_2\text{O}} \chi_{\text{H}_2\text{O}} \delta_{\text{CO}_2, \text{H}_2\text{O}}) - Z = 0, \quad (20)$$

385 where the coefficients c_0 – c_3 in F_2 depend on Z through $\delta = g(\eta)\kappa [\exp(\varepsilon/k_B T) -$
 386 $1]$, $\delta_{\text{CO}_2, \text{H}_2\text{O}} = S_{\text{CO}_2, \text{H}_2\text{O}} \delta$ is the scaled cross-association strength, and A , B
 387 are the dimensionless SRK parameters defined in Section 2. Equation (18)
 388 is the residual of the compressibility equation, (19) is the cubic equation
 389 for the H₂O bonding fraction $\chi_{\text{H}_2\text{O}}$ (four sites, 4C scheme), and (20) is the
 390 algebraic closure for the CO₂ cross-association fraction χ_{CO_2} . These three
 391 equations are solved simultaneously by Newton–Raphson, using the solution
 392 from the previous SSI iteration as the initial guess. The 3×3 Jacobian is
 393 derived analytically; all partial derivatives of F_1 , F_2 , and F_3 with respect to
 394 $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ are available in closed form via the chain rule through $g(\eta)$
 395 and its first two derivatives (Section S3.1 of the Supplemental Information).
 396 A scan+Brent fallback is retained for robustness when no warm start is avail-
 397 able. Crucially, the same system F_1 – F_3 is applied to *both* phases using their
 398 respective compositions and SRK parameters; the two phases differ only in
 399 input, not in structure. This phase-symmetry breaks in eCPA: the aqueous

400 phase requires a structurally different system that additionally couples Z^{aq}
 401 to the relative permittivity ε_r , as described in the next Section.

402 3.4. K -value flash for the eCPA ternary system

403 In the full eCPA ternary ($\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$), the equilibrium NaCl
 404 molality in the aqueous phase m_s^{aq} couples to the flash equations through
 405 the aqueous-phase fugacity coefficients. Since NaCl is non-volatile, all salt
 406 remains in the aqueous phase and m_s^{aq} is fully determined by the current
 407 K -values and the feed composition — it carries no independent degree of
 408 freedom. We exploit this to *eliminate* m_s^{aq} analytically within each SSI step:
 409 given the current $(K_{\text{H}_2\text{O}}, K_{\text{CO}_2})$, the compositions of both phases and the
 410 molality are computed algebraically before the fugacity-coefficient evaluation,
 411 so the entire flash reduces to a single-level SSI on $(K_{\text{H}_2\text{O}}, K_{\text{CO}_2})$ with no outer
 412 loop.

413 *Analytical elimination of m_s^{aq} .* Given $K_{\text{H}_2\text{O}} \equiv y_{\text{H}_2\text{O}}/x_{\text{H}_2\text{O}}$ and $K_{\text{CO}_2} \equiv y_{\text{CO}_2}/x_{\text{CO}_2}$,
 414 the compositions of both phases are fixed once $x_{\text{H}_2\text{O}}$ is known. On a one-mole
 415 volatile-feed basis with CO_2 feed fraction z_{CO_2} (so $z_{\text{H}_2\text{O}} = 1 - z_{\text{CO}_2}$ is the
 416 H_2O feed fraction) and feed molality m_s , the H_2O molar balance relates β
 417 to $x_{\text{H}_2\text{O}}$: $\beta(x_{\text{H}_2\text{O}}) = [z_{\text{H}_2\text{O}}/x_{\text{H}_2\text{O}} - 1] / (K_{\text{H}_2\text{O}} - 1)$. The CO_2 -rich phase nor-
 418 malization ($y_{\text{H}_2\text{O}} + y_{\text{CO}_2} = 1$, no NaCl) gives $x_{\text{CO}_2} = (1 - K_{\text{H}_2\text{O}} x_{\text{H}_2\text{O}})/K_{\text{CO}_2}$.
 419 Substituting $x_2 = x_3 = x_{\text{H}_2\text{O}} m_s^{\text{aq}} M_w$ and the NaCl molar balance $m_s^{\text{aq}} =$
 420 $m_s z_{\text{H}_2\text{O}} / [(1 - \beta) x_{\text{H}_2\text{O}}]$ into the aqueous normalization $\sum_i x_i = 1$ yields a
 421 single scalar equation in $x_{\text{H}_2\text{O}}$:

$$x_{\text{H}_2\text{O}} + \frac{2 m_s z_{\text{H}_2\text{O}} M_w}{1 - \beta(x_{\text{H}_2\text{O}})} + \frac{1 - K_{\text{H}_2\text{O}} x_{\text{H}_2\text{O}}}{K_{\text{CO}_2}} = 1. \quad (21)$$

422 Equation (21) is monotone in $x_{\text{H}_2\text{O}}$ over the physical domain and is solved
 423 by Brent's method at each SSI step. Because this is a purely algebraic equa-
 424 tion (no EoS evaluations), the scalar root-find adds negligible cost relative
 425 to the fugacity-coefficient evaluations that follow. Once $x_{\text{H}_2\text{O}}$ is known, all
 426 compositions and the molality follow analytically:

$$m_s^{\text{aq}} = \frac{m_s z_{\text{H}_2\text{O}}}{[1 - \beta(x_{\text{H}_2\text{O}})] x_{\text{H}_2\text{O}}}. \quad (22)$$

427 *K -value iteration and acceleration.* With compositions fully determined by
 428 the current $(K_{\text{H}_2\text{O}}, K_{\text{CO}_2})$, the fugacity residual is a 2-vector

$$g_i^{(k)} = \ln \hat{\phi}_i^c(y_{\text{H}_2\text{O}}^{(k)}) + \ln y_i^{(k)} - \ln \hat{\phi}_i^{\text{aq}}(x_{\text{H}_2\text{O}}^{(k)}, m_s^{\text{aq},(k)}) - \ln x_i^{(k)}, \quad i \in \{\text{H}_2\text{O}, \text{CO}_2\}, \quad (23)$$

429 and the K-value update follows Eq. (14) with the Jex acceleration (Eq. (15)).

430 *Inner Newton–Raphson solve for Z^{aq} , ε_r , and $\chi_{\text{H}_2\text{O}}$ (eCPA).* Within each SSI
 431 iteration, the eCPA aqueous-phase fugacity coefficients require simultaneous
 432 solution of Z^{aq} , the relative permittivity ε_r , and the H_2O association frac-
 433 tion $\chi_{\text{H}_2\text{O}}$. The permittivity is coupled to the association state through the
 434 Onsager–Kirkwood self-consistency equation [7], and both feed back into the
 435 compressibility through the Debye–Hückel and permittivity correction terms
 436 absent in the salt-free CPA. The CO_2 cross-association fraction is eliminated
 437 algebraically, $\chi_{\text{CO}_2} = Z^{\text{aq}}/D_c$ with $D_c = Z^{\text{aq}} + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w$, re-
 438 ducing the aqueous inner problem to three coupled equations:

$$G_0 = Z^{\text{aq}} - \frac{Z^{\text{aq}}}{Z^{\text{aq}} - B^{\text{aq}}} + \frac{A^{\text{aq}}}{Z^{\text{aq}} + B^{\text{aq}}} - Z^{\text{aq,DH}} - Z^{\text{aq,Perm}} - \left(1 + \frac{\eta}{g_w} \frac{dg_w}{d\eta}\right) [x_{\text{H}_2\text{O}}(\chi_{\text{H}_2\text{O}} - 1) + x_{\text{CO}_2}(\chi_{\text{CO}_2} - 1)] = 0, \quad (24)$$

$$G_1 = \frac{(\varepsilon_r - \varepsilon_\infty)(2\varepsilon_r + \varepsilon_\infty)}{\varepsilon_r(\varepsilon_\infty + 2)^2} - \frac{N_A \rho_N \mu_{0w}^2}{9\varepsilon_0 k_B T} x_{\text{H}_2\text{O}} g_w(\chi_{\text{H}_2\text{O}}) = 0, \quad (25)$$

$$G_2 = \chi_{\text{H}_2\text{O}} - \frac{Z^{\text{aq}}}{Z^{\text{aq}} + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}\Delta_w + 2x_{\text{CO}_2}\chi_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w} = 0. \quad (26)$$

439 Equation (24) is the eCPA compressibility residual: the SRK physical term
 440 is augmented by the Wertheim association correction (as in CPA) and by the
 441 Debye–Hückel compressibility $Z^{\text{aq,DH}}$ and the permittivity correction $Z^{\text{aq,Perm}}$
 442 [9], both of which depend on ε_r . Equation (25) is the Onsager–Kirkwood
 443 self-consistency condition: its left-hand side is the Onsager dielectric func-
 444 tion of ε_r and $\varepsilon_\infty(Z^{\text{aq}})$ (the high-frequency permittivity from the Clausius–
 445 Mossotti relation), while the right-hand side couples the total mixture molar
 446 density $\rho_N = P/(Z^{\text{aq}}RT)$ and H_2O mole fraction $x_{\text{H}_2\text{O}}$ to the Kirkwood
 447 g-factor $g_w(\chi_{\text{H}_2\text{O}})$ through the permanent dipole moment μ_{0w} [9]. Equa-
 448 tion (26) is the self-consistency condition for the H_2O bonding fraction, with
 449 $\Delta_w = g_w(\eta) \kappa_w [\exp(\varepsilon_w/k_B T) - 1]$ the association strength. These three equa-
 450 tions are solved simultaneously by Newton–Raphson, warm-started from the
 451 previous SSI iterate. The 3×3 Jacobian $\mathbf{H} = \partial(G_0, G_1, G_2)/\partial(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$
 452 is derived analytically in Section S3.2 of the Supplemental Information. The
 453 CO_2 -rich phase retains the 3×3 system of Section 3.3 (with $x_{\text{H}_2\text{O}}^c = 0$ elimi-
 454 nating the eCPA terms); when no warm start is available, a general-purpose
 455 root-finder provides the cold-start fallback for the aqueous phase. Conver-
 456 gence is declared when $\|\mathbf{G}^{(k)}\| < 10^{-8}$.

457 In the hierarchical algorithm described in Section 3.6, K-values are al-
 458 ways initialized from the stability test result, with Wilson’s correlation as a
 459 fallback. When the flash is invoked without a prior stability result, four fixed
 460 ($K_{\text{H}_2\text{O}}$, K_{CO_2}) seeds are tried in sequence until one converges.

461 *Outer Newton polish for the eCPA K-value flash.* By analogy with the CPA
 462 flash (“Newton polish” paragraph in Section 3.3), we test an outer Newton–
 463 Raphson polish applied once the K-value SSI residual $\|\mathbf{g}\|$ falls below a
 464 switchover threshold. Because the eCPA K-value residual couples the highly
 465 non-linear electrostatic and association contributions through the implicitly
 466 defined aqueous composition $x_{\text{H}_2\text{O}}$ (Eq. (21)), deriving an analytical 2×2
 467 Jacobian $\partial \mathbf{g} / \partial \ln \mathbf{K}$ would require implicit differentiation through a scalar
 468 Brent solve at every Newton step. We therefore test the approach with a
 469 forward-difference numerical Jacobian, which adds two additional $\ln \phi$ evalu-
 470 ations per Newton step. The benchmarks (672 conditions spanning $T = 37$ –
 471 157°C , $P = 60$ – 400 bar, $z_{\text{CO}_2} = 0.15$ – 0.70 , $m_s = 0.5$ – 4.5 mol kg $^{-1}$) show that
 472 on the warm-start path — the production path in sequential simulation —
 473 the SSI already converges in approximately one iteration because the table
 474 initial guess is close to the solution, so Newton is never triggered and the
 475 overhead is negligible. On the cold-start path, where SSI requires ~ 7 iter-
 476 ations on average, the Newton polish reduces the total iteration count by
 477 25% (from 7.15 to 5.35 for a switchover threshold of 10^{-3} , or 17% for 10^{-4}),
 478 but the numerical-Jacobian overhead — three $\ln \phi$ evaluations per Newton
 479 step versus one for SSI — cancels the iteration savings in wall time ($\lesssim 1\%$
 480 improvement). Because the warm-start path dominates in practice and a
 481 worthwhile wall-time speedup would require an analytical Jacobian that is
 482 non-trivial to derive, the outer Newton polish is not applied by default in the
 483 eCPA flash.

484 3.5. Phase envelope

485 We trace the two-phase boundary (P – T locus at fixed z_{CO_2} and m_s) using
 486 the following procedure:

- 487 1. Scan a coarse grid of (T, P) points with the stability test to classify
 488 each cell as stable or unstable.
- 489 2. Identify transitions between stable and unstable cells.
- 490 3. Refine each transition to the phase boundary by bisection, using SSI
 491 to solve the flash at each trial point.

492 To handle the large number of (T, P) combinations efficiently, the bisection
 493 at each boundary point is executed in parallel across all available CPU cores.

494 3.6. Fast flash for reservoir simulation

495 3.6.1. Three-dimensional solution table

496 We precompute flash solutions on a three-dimensional grid (T, P, m_s)
 497 with:

- 498 • T : 361 points from 0 to 360 °C (1 °C spacing), covering conditions from
 499 cold-basin CO₂ storage to hydrothermal and geothermal applications.
- 500 • $\log_{10} P$: 100 points from $\log_{10}(1)$ to $\log_{10}(2000)$ (log-uniform).
- 501 • m_s (NaCl molality): 14 points from 0.0 to 6.0 mol kg⁻¹, spanning the
 502 full range of experimental data from dilute brine to near-saturation.

503 This gives $361 \times 100 \times 14 = 505,400$ grid cells. Note that m_s (no superscript)
 504 is the feed molality, which differs from the equilibrium aqueous molality
 505 $m_s^{\text{aq}} \geq m_s$ due to the salting-out effect. The overall CO₂ feed fraction z_{CO_2}
 506 is not included as a table dimension. All table entries are computed at a
 507 representative feed fraction $z_{\text{CO}_2} = 0.5$; the equilibrium quantities stored
 508 (K-values, Z , χ , m_s^{aq}) depend weakly on z_{CO_2} for fixed (T, P, m_s) . This
 509 dependence is primarily through the salting-out effect: a larger CO₂ feed
 510 fraction partitions more water into the CO₂-rich phase, concentrating the
 511 brine and raising m_s^{aq} , which in turn slightly shifts the K-values. We verify
 512 numerically that at typical subsurface conditions ($T = 50$ °C, $P = 100$ bar,
 513 $m_s = 1$ mol kg⁻¹) the spread of K-values across the full two-phase z_{CO_2} range
 514 is less than 0.15%, confirming that the table provides adequate warm-start
 515 quality across the full feed composition range. Excluding z_{CO_2} reduces the
 516 table size by more than an order of magnitude relative to a four-dimensional
 517 grid.

518 Each cell is solved using the full hierarchical algorithm (Section 3.2): the
 519 Michelsen stability test (six trial compositions) is performed first; only gen-
 520 uinely unstable conditions proceed to K-value SSI flash. This means no exter-
 521 nal assumption about the phase state is required; every cell is self-consistently
 522 classified and solved. At each two-phase cell we store the converged K-values
 523 ($K_{\text{H}_2\text{O}}, K_{\text{CO}_2}$), the inner-Newton state for the aqueous phase ($Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}}$)
 524 and for the CO₂-rich phase (Z^c, χ_{CO_2}), and the equilibrium aqueous molality

525 m_s^{aq} . All seven quantities serve directly as warm-start inputs to the hierar-
 526 chical flash: K-values initialize the outer SSI, while Z , ε_r , and χ initialize
 527 the inner Newton solves for compressibility factor and association fractions
 528 (Section S3.2 of the Supplemental Information), eliminating the need for cold-
 529 start Newton iterations. A scalar stability flag (two-phase or single-phase) is
 530 also stored.

531 Table lookup uses a trilinear interpolant operating in $(T, \log_{10} P, m_s)$
 532 space, which is vectorized and evaluates in <1 ms.

533 3.6.2. Fast flash algorithm

534 The fast flash routine operates as follows. A query to the 3D interpolant
 535 in $(T, \log_{10} P, m_s)$ yields an initial state vector (compressibility factors, phase
 536 compositions, association fractions, and permittivity) together with a stabil-
 537 ity classification (two-phase or single-phase). If the interpolant classifies the
 538 condition as two-phase, the Michelsen stability test is skipped and SSI is ini-
 539 tialized directly from the table state vector with no damping ($\omega = 1.0$). If the
 540 classification is single-phase or uncertain (for example, near the two-phase
 541 boundary or for grid cells with low convergence quality), the full Michelsen
 542 TPD test is performed; a stable result returns immediately without a flash,
 543 while an unstable result proceeds to SSI. In either two-phase path, the warm-
 544 started SSI converges in 3–4 iterations because the interpolated state vector
 545 is already within $\sim 1\%$ of the converged solution. Should SSI fail to converge,
 546 a full cold-start with Wilson K-value initialization and damped SSI ($\omega = 0.7$)
 547 provides a robust fallback.

548 The key insight enabling this strategy is that the solution surface — the
 549 mapping from (T, P, m_s) to the equilibrium state vector — is smooth and
 550 slowly varying except near the two-phase boundary and the CO_2 critical point
 551 ($T_{c,\text{CO}_2} = 31.3^\circ\text{C}$, $P_c = 73.8$ bar). A trilinear interpolant from a grid with
 552 1°C temperature spacing and log-uniform pressure spacing (100 points over
 553 1–2000 bar) therefore provides an excellent initial guess for the vast majority
 554 of reservoir conditions. The selective stability check further reduces cost: in
 555 a typical reservoir simulation, the majority of grid cells are well within either
 556 a single-phase or a two-phase region (injected CO_2 coexisting with brine) and
 557 the interpolant stability flag can be trusted, while the expensive Michelsen
 558 test is reserved for cells near boundaries where phase transitions occur.

559 *3.7. Alternative simplified flash exploiting $y_{\text{H}_2\text{O}} \approx 0$*

560 At typical CO_2 storage reservoir conditions ($T \lesssim 100^\circ\text{C}$), the H_2O con-
 561 tent of the CO_2 -rich phase remains below 1 mol% (see Fig. 8 and the dis-
 562 cussion in Section 5.1.2). The approximation $y_{\text{H}_2\text{O}} = 0$ then collapses the
 563 standard two-component K-value flash to a single scalar equation, enabling
 564 a simpler, self-contained calculation without the outer salinity iteration.

565 *Mathematical basis.* Under $y_{\text{H}_2\text{O}} = 0$ the CO_2 -rich phase is pure CO_2 , so its
 566 fugacity reduces to the pure-component value:

$$f_{\text{CO}_2}^c(T, P) = P \phi_{\text{CO}_2}^{\text{pure}}(T, P). \quad (27)$$

567 A compact material-balance argument shows that the aqueous NaCl mo-
 568 lality equals the feed value exactly under this assumption. Define the ef-
 569 fective H_2O feed mole fraction $z_{\text{H}_2\text{O}} = (1 - z_{\text{CO}_2})/(1 + 2m_s M_w)$, where
 570 $M_w = 0.018015 \text{ kg mol}^{-1}$. From the NaCl and H_2O balances (salt is non-
 571 volatile, so $z_{\text{NaCl}} = (1 - \beta) x_{\text{H}_2\text{O}}^{\text{aq}} m_s^{\text{aq}} M_w$ and $z_{\text{H}_2\text{O}} = (1 - \beta) x_{\text{H}_2\text{O}}^{\text{aq}}$), dividing
 572 gives $m_s^{\text{aq}} = z_{\text{NaCl}}/(z_{\text{H}_2\text{O}} M_w) = m_s$. The equilibrium is therefore determined
 573 by a single equation in $x_{\text{H}_2\text{O}}^{\text{aq}}$:

$$f_{\text{CO}_2}^{\text{aq}}(x_{\text{H}_2\text{O}}^{\text{aq}}, m_s, T, P) = f_{\text{CO}_2}^c(T, P), \quad (28)$$

574 where $x_{\text{CO}_2}^{\text{aq}} = 1 - x_{\text{H}_2\text{O}}^{\text{aq}}(1 + 2m_s M_w)$ and the aqueous fugacity is evaluated
 575 with the eCPA 3×3 Newton solver (Section 3.4). The phase fraction follows
 576 from the H_2O material balance:

$$\beta = 1 - \frac{z_{\text{H}_2\text{O}}}{x_{\text{H}_2\text{O}}^{\text{aq}}}. \quad (29)$$

577 Phase identification is embedded in the root-finding step: if the CO_2 -fugacity
 578 residual evaluated at $x_{\text{H}_2\text{O}}^{\text{aq}} = z_{\text{H}_2\text{O}}$ (i.e. $\beta = 0$) is non-positive, the mixture is
 579 single-phase aqueous and no further solve is needed.

580 *Algorithm.* The pure- CO_2 fugacity (27) is computed once via the CO_2 -rich-
 581 phase inner Newton solver at $x_{\text{H}_2\text{O}}^c = 0$. Equation (28) is then solved by
 582 Brent’s method over the bracket $[z_{\text{H}_2\text{O}}, x_{\text{H}_2\text{O}}^{\text{max}}]$ where $x_{\text{H}_2\text{O}}^{\text{max}} = 1/(1 + 2m_s M_w)$
 583 is the H_2O mole fraction of a salt-only aqueous phase. Warm-starting of the
 584 inner Newton solver across Brent evaluations (carrying forward $[Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}}]$
 585 from each accepted step) reduces the inner-solve cost to the same 3–6 Newton
 586 iterations achieved in the full flash. The formulation is valid for both the
 587 salt-free CPA binary ($m_s = 0$) and the full eCPA ternary; no outer salinity
 588 iteration is required in either case. Results are assessed against the full K-
 589 value flash in Section 5.3.

590 4. Validation

591 4.1. $\text{CO}_2 + \text{H}_2\text{O}$ binary

592 We validate both the (new and tuned) salt-free CPA EOS and the eCPA
 593 EOS at $m_s = 10^{-4} \text{ mol kg}^{-1}$ (effectively salt-free) against 631 experimental
 594 VLE data points compiled by Coelho et al. [9] for $T = 0\text{--}350^\circ\text{C}$, achieving
 595 100% convergence across all conditions. Both phases are characterised: the
 596 CO_2 mole fraction in the aqueous phase x_{CO_2} and the H_2O mole fraction in
 597 the CO_2 -rich phase $y_{\text{H}_2\text{O}}$.

598 Figure 1 shows representative comparisons at three temperatures span-
 599 ning the validated range ($T = 25, 150, \text{ and } 250^\circ\text{C}$); results for the remaining
 600 32 temperatures are collected in Fig. S1 of the Supplemental Information
 601 (phase split results for molalities up to $m_s = 6 \text{ mol kg}^{-1}$ are also included for
 602 comparison and discussed further below). Table 1 summarizes the accuracy
 603 metrics by condition regime. CPA and eCPA give essentially identical re-
 604 sults for the salt-free binary (an encouraging consistency validation between
 605 independent codes), so only eCPA values are reported. The results in Fig. 1
 606 use a feed composition of 50 mol% CO_2 and the full stability+flash frame-
 607 work; Coelho et al. [9] report VLE compositions only (phase fractions are
 608 not needed for their parameterization validation).

609 Over all 631 conditions, the overall AARE is 9.6% for x_{CO_2} ($N = 560$)
 610 and 24.4% for $y_{\text{H}_2\text{O}}$ ($N = 452$); the larger error for the CO_2 -rich phase is
 611 consistent with the Coelho et al. [9] parametrization, which prioritizes brine
 612 conditions. However, these headline numbers are inflated by three challenging
 613 regimes:

- 614 1. *Near the CO_2 critical point* ($T_{c,\text{CO}_2} = 31.3^\circ\text{C}$, $P_c = 73.8 \text{ bar}$): K-value
 615 gradients with pressure are steep and small errors in fugacity coeffi-
 616 cients translate to large errors in solubility, consistent with the general
 617 difficulty of describing near-critical behavior with equations of state
 618 based on a cubic repulsive term.
- 619 2. *Near the H_2O critical point* ($T \geq 260^\circ\text{C}$): the two-phase envelope
 620 narrows and the mixture critical locus predicted by the EoS deviates
 621 slightly from experiment. At $T = 268\text{--}270^\circ\text{C}$, experimental composi-
 622 tions become identical ($K_i = 1$), indicating conditions at or beyond the
 623 mixture critical point.
- 624 3. *Ultra-high pressures* ($P > 1500 \text{ bar}$): beyond typical reservoir depths
 625 and outside the solution-table range. The sole data source at $P >$

2000 bar [23] cannot be independently cross-validated, and errors in this regime reach 78% AARE for $y_{\text{H}_2\text{O}}$.

4. *Near-freezing conditions* ($T \lesssim 12^\circ\text{C}$): these conditions overlap with the CO_2 hydrate stability zone, where solid CO_2 clathrates can form in preference to a liquid–vapor equilibrium. The eCPA EoS is not parameterized for hydrate formation and computes fluid-phase VLE only; experimental data in this regime may therefore reflect conditions where hydrate formation partially controls the observed CO_2 solubility.

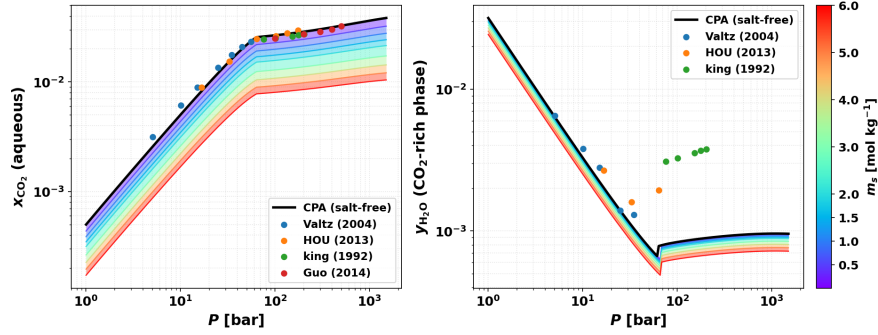
For conditions most relevant to subsurface CO_2 storage ($T = 30\text{--}150^\circ\text{C}$, $P = 50\text{--}600$ bar), the AARE improves to **6.5%** for x_{CO_2} ($N = 137$).

A cross-reference consistency analysis (Fig. S7) further reveals that the $y_{\text{H}_2\text{O}}$ data of King et al. [24] are physically suspect: the reported water content in the CO_2 -rich phase *increases* with pressure above the CO_2 saturation pressure, contradicting thermodynamic expectations and the measurements of Meyer and Harvey [25] at overlapping conditions. After excluding flagged outlier references (92 of 631 points, see Fig. S7), the overall $y_{\text{H}_2\text{O}}$ AARE decreases from 24.4% to 17.4%.

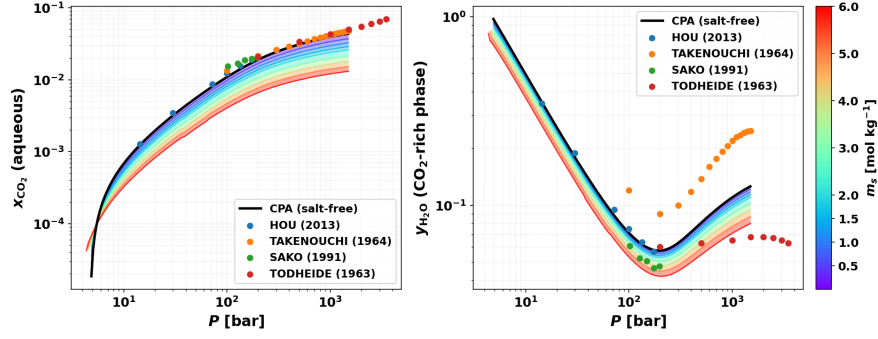
Table 1: VLE accuracy metrics for the $\text{CO}_2 + \text{H}_2\text{O}$ binary by condition regime. AARE: average absolute relative error; AAE: average absolute error (mol%). CPA and eCPA ($m_s = 10^{-4}$) give identical results for this salt-free system; only eCPA values are shown. “Excl. outliers” removes references flagged by cross-reference consistency (see text).

Regime	x_{CO_2} (aqueous)			$y_{\text{H}_2\text{O}}$ (CO_2 -rich)		
	N	AARE (%)	AAE (mol%)	N	AARE (%)	AAE (mol%)
All data	560	9.6	0.53	452	24.4	3.5
All (excl. outliers)	468	9.3	0.53	355	17.4	3.0
Subsurface	137	6.5	0.13	102	25.6	0.68
Extended ($T = 10\text{--}200^\circ\text{C}$)	362	8.9	0.21	299	24.6	1.7
High- T ($T \geq 260^\circ\text{C}$)	129	10.1	1.6	111	15.9	8.0
High- P ($P > 1500$ bar)	27	18.4	2.9	27	78.0	15.0

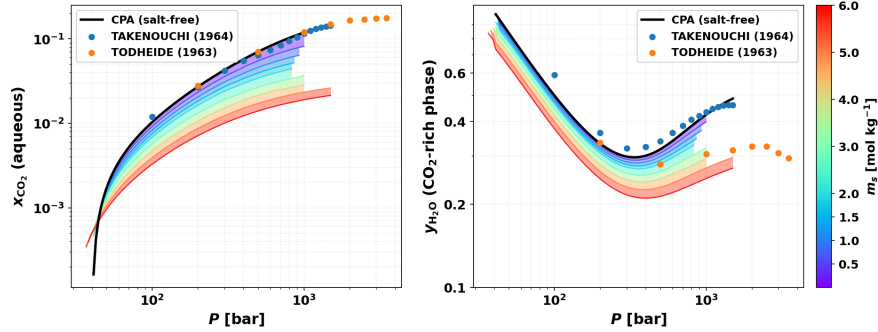
Subsurface: $T = 30\text{--}150^\circ\text{C}$, $P = 50\text{--}600$ bar.



(a) 25 °C (298 K)



(b) 150 °C (423 K)



(c) 250 °C (523 K)

Figure 1: $\text{CO}_2 + \text{H}_2\text{O}$ phase equilibria at three representative temperatures (25, 150, and 250 °C). Each panel shows the CO_2 mole fraction in the aqueous phase (x_{CO_2} , left sub-panel) and the H_2O mole fraction in the CO_2 -rich phase ($y_{\text{H}_2\text{O}}$, right sub-panel) vs. pressure. Open symbols: experimental data [9]; black solid line: CPA (salt-free); rainbow-colored filled bands: eCPA predictions spanning NaCl molalities $m_s = 0\text{--}6 \text{ mol kg}^{-1}$ (filled between consecutive molality contours; violet ≈ 0 , red = 6 mol kg^{-1}), with discrete eCPA curves overlaid. Results for the remaining 32 temperatures are shown in Fig. S1.

643 These model errors should be interpreted in the context of the experi-
644 mental uncertainty in the compiled dataset. Two independent estimates of
645 measurement scatter can be obtained directly from the data. First, at the 76
646 (T, P) conditions where at least two independent references overlap (matched
647 within ± 2.5 bar), the coefficient of variation between reported values aver-
648 ages 4.0% for x_{CO_2} and 11.1% for $y_{\text{H}_2\text{O}}$. Second, the Gasser–Sroka–Jennen-
649 Steinmetz (GSJS) difference-based variance estimator [26], which estimates
650 measurement noise from the scatter of consecutive data points around a lo-
651 cally interpolated trend within each isothermal series, yields a median relative
652 noise of 2.4% for x_{CO_2} and 4.7% for $y_{\text{H}_2\text{O}}$. Taken together, these estimates
653 suggest an intrinsic experimental uncertainty of roughly 2–4% for x_{CO_2} and
654 5–11% for $y_{\text{H}_2\text{O}}$, so the model AARE of 6.5% (subsurface) to 9.6% (global)
655 for x_{CO_2} are close to the measurement noise.

656 Moreover, relative errors can be misleading when mole fractions span
657 several orders of magnitude: a 35% AARE for $y_{\text{H}_2\text{O}} < 0.005$ (101 points)
658 corresponds to an average absolute error (AAE) of only 0.1 mol%—negligible
659 for reservoir simulation. Across all data, the AAE is 0.53 mol% for x_{CO_2} and
660 3.5 mol% for $y_{\text{H}_2\text{O}}$; under subsurface conditions these drop to 0.13 and 0.68
661 mol%, respectively.

662 4.2. $\text{CO}_2 + \text{NaCl}$ brine

663 Figure 1 also illustrates the effect of brine salinity on the equilibrium
664 compositions by including predictions for the full molality range considered
665 in this work. We further validate the eCPA ternary flash against all available
666 experimental data from the literature [27, 28, 34, 29, 33, 35, 36, 37, 38, 39,
667 31, 30, 32, 40, 41], spanning $T = 5\text{--}450^\circ\text{C}$, $P = 1\text{--}1400$ bar, and $m_s = 0.17\text{--}$
668 6.0 mol kg^{-1} NaCl (708 experimental data points in total). Two types of
669 measurements are compared: CO_2 solubility in the aqueous phase and the
670 CO_2 mole fraction in the CO_2 -rich phase (y_{CO_2}). Aqueous-phase solubility is
671 reported differently by different sources: as CO_2 molality (m_{CO_2} [mol kg $^{-1}$]),
672 or as the mole fraction on a NaCl-free basis ($x_{\text{CO}_2}^{\text{sf}} = x_{\text{CO}_2}/(x_{\text{CO}_2} + x_{\text{H}_2\text{O}})$),
673 where NaCl is excluded from the denominator. The latter is a reporting
674 convention used by some authors and describes the same physical quantity
675 as m_{CO_2} ; it is converted to molality via $m_{\text{CO}_2} = x_{\text{CO}_2}^{\text{sf}}/[(1 - x_{\text{CO}_2}^{\text{sf}})M_w]$ before
676 comparison.

677 All experimental results and eCPA predictions are illustrated in Figs. 2,
678 3, S2, and S3.

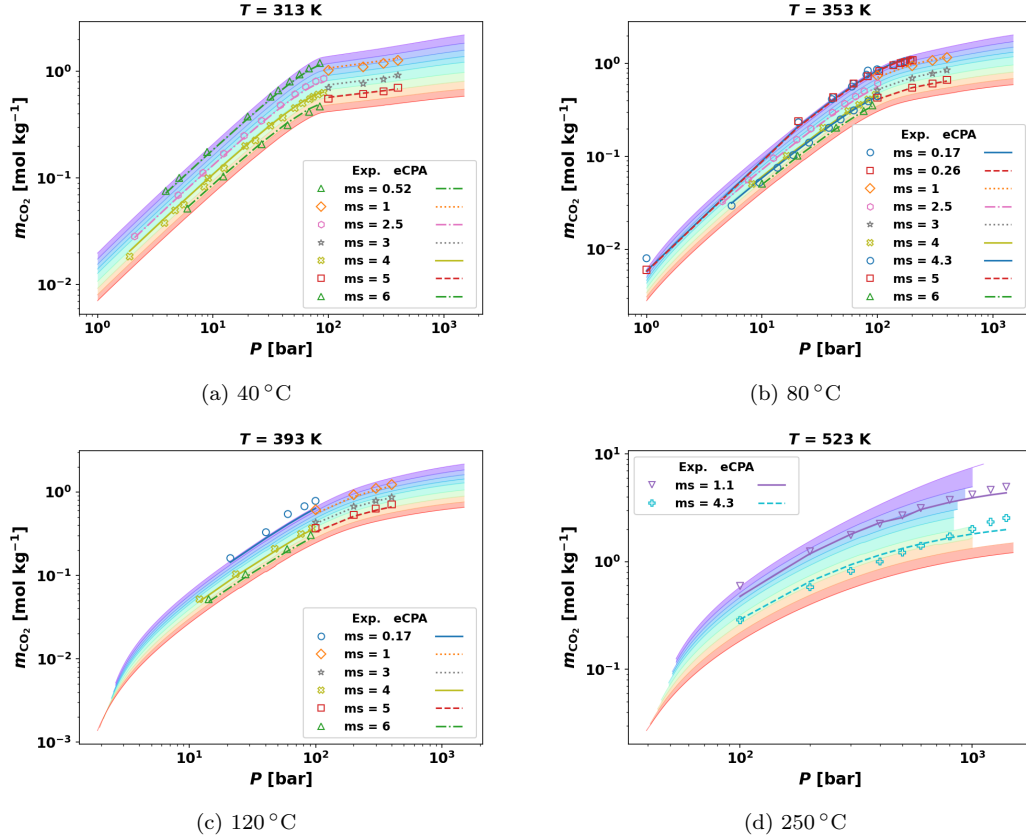


Figure 2: CO₂ solubility in NaCl brine at four representative temperatures (40, 80, 120, and 250 °C). Each panel shows CO₂ molality m_{CO_2} [mol kg⁻¹] vs. pressure (log-log). Open symbols: experimental data [27, 28, 29, 30, 31, 32, 33]; discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions spanning $m_s = 0\text{--}6$ mol kg⁻¹ (color scale as in Fig. 1: violet ≈ 0 , red = 6 mol kg⁻¹). Temperatures with additional CO₂-rich phase data (y_{CO_2}) are shown separately in Fig. 3; all 18 temperatures in Figs. S2 and S3.

Table 2 shows the accuracy breakdown. 689 of 708 points converge to a two-phase solution using the K-value SSI warm-started from the solution table; the model predicts single-phase behavior at 18 points (particularly at $T \geq 350$ °C where CO₂ + H₂O approaches complete miscibility in the model) and fails to converge at one extreme high- P , high- m_s condition, giving an overall two-phase convergence rate of 97.3%. The K-value SSI requires a median of 6 outer iterations per converged call. Overall AARE across all converged points is 6.2%, with a slight underprediction bias of -1.5% . The

687 AARE for m_{CO_2} across all T and m_s is 6.6%, in excellent agreement with the
 688 6.9% reported by Coelho et al. [9] using the original eCPA parameterization.
 689 The CO_2 -rich phase composition y_{CO_2} is reproduced to within 0.4% AARE,
 690 reflecting the near-purity of the CO_2 -rich phase and its weak dependence on
 691 salinity.

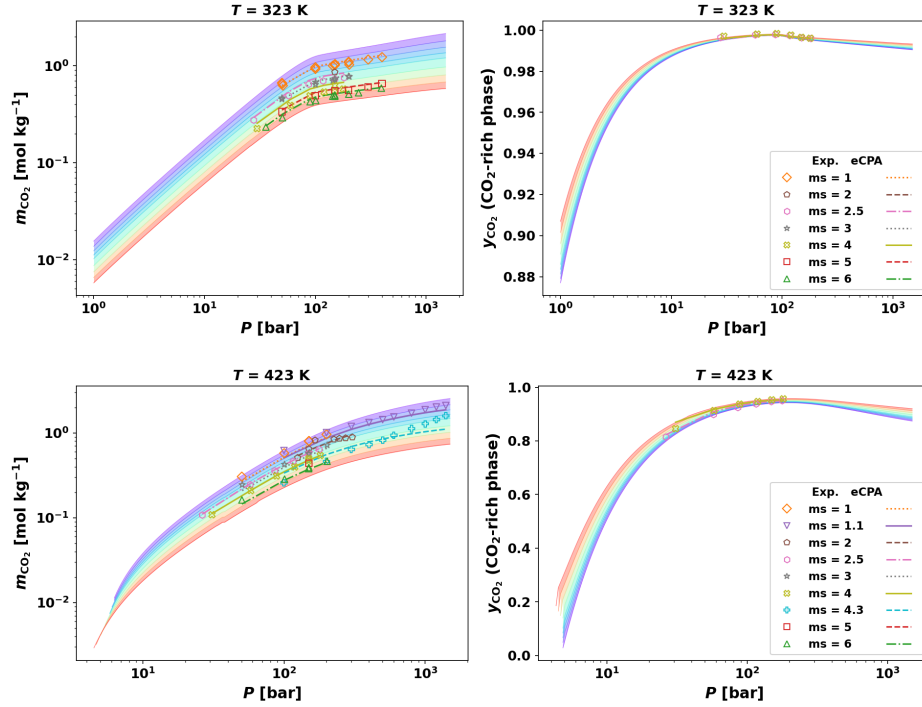


Figure 3: $\text{CO}_2 + \text{NaCl}$ brine phase equilibria at temperatures for which both CO_2 molality (m_{CO_2} , left panel) and CO_2 mole fraction in the CO_2 -rich phase (y_{CO_2} , right panel) data are available: (top) 50°C , (bottom) 150°C (see also Fig. S3 for 100°C). Open symbols: experimental data [31, 35, 27, 37, 34, 32, 36, 41, 33, 38, 39]; discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions across $m_s = 0\text{--}6 \text{ mol kg}^{-1}$ (color scale as in Fig. 1).

692 At hydrothermal temperatures, AARE for m_{CO_2} rises to 14.0% at $T =$
 693 300°C and 12.8% at $T = 350^\circ\text{C}$, reflecting reduced eCPA accuracy at condi-
 694 tions beyond the conventional reservoir temperature range. At $T \geq 400^\circ\text{C}$,
 695 the available experimental pressures (300–900 bar, all from Takenouchi and
 696 Kennedy 33) fall in the region where eCPA predicts single-phase behavior,
 697 suggesting that the $\text{CO}_2 + \text{H}_2\text{O}$ miscibility gap closes at these extreme con-
 698 ditions in the model. Notably, accuracy at the lowest temperatures ($T = 5\text{--}$

10 °C) is also reduced (AARE 14–21%), reflecting the challenge of modeling CO₂ solubility very close to the freezing point where eCPA is not specifically parameterized. These conditions furthermore fall within the CO₂ hydrate stability zone, where solid CO₂ clathrates can form; the eCPA EoS computes fluid-phase VLE only and does not model hydrate formation explicitly. Fig. S4 provides a comprehensive view of the error distribution across the full (T, m_s) parameter space: accuracy is best at intermediate temperatures ($T = 27\text{--}227^\circ\text{C}$) and moderate salinities ($m_s \leq 3 \text{ mol kg}^{-1}$), and degrades systematically at the boundaries of the validated range.

Table 2: VLE accuracy metrics for CO₂ + NaCl brine ($T = 5\text{--}450^\circ\text{C}$, $m_s = 0.17\text{--}6.0 \text{ mol kg}^{-1}$, 708 data points total, 689 two-phase converged). Bias = $\langle(\hat{y} - y)/y\rangle$.

Quantity	N	AARE (%)	Bias (%)
$m_{\text{CO}_2} [\text{mol kg}^{-1}]$	501	6.6	−3.3
$x_{\text{CO}_2}^{\text{sf}}$ (NaCl-free basis, conv. to m_{CO_2})	116	5.5	+2.7
x_{CO_2} [salt-incl.]	36	8.6	+8.6
y_{CO_2}	36	0.5	−0.1
All converged	689	6.2	−1.5

4.3. Flash results as a function of feed composition

A central motivation for full stability+flash — rather than straightforward VLE calculations at a fixed salinity — arises naturally in reservoir simulation. When CO₂ is injected into a saline aquifer, the reservoir fluid at any given grid cell can be represented as a mixture of brine (carrying NaCl at feed molality m_s) and varying amounts of injected CO₂. The overall CO₂ mole fraction z_{CO_2} in this mixture is the primary degree of freedom, and stability+flash must be performed to determine the equilibrium state.

As shown in Section 1, in the ternary system the equilibrium molality m_s^{aq} exceeds the feed value because NaCl is concentrated as H₂O partitions into the CO₂-rich phase; quantitatively (from Eq. 22):

$$m_s^{\text{aq}} = m_s \frac{(1 - \beta) x_{\text{H}_2\text{O}} + \beta y_{\text{H}_2\text{O}}}{(1 - \beta) x_{\text{H}_2\text{O}}} = m_s \left(1 + \frac{\beta y_{\text{H}_2\text{O}}}{(1 - \beta) x_{\text{H}_2\text{O}}} \right),$$

where β is the CO₂-rich molar phase fraction. Because $\beta y_{\text{H}_2\text{O}} > 0$ in the two-phase region, m_s^{aq} exceeds m_s , and the excess grows with z_{CO_2} as β increases.

721 This elevated salinity modifies the aqueous-phase fugacity coefficients and in
 722 turn shifts the equilibrium phase compositions — an effect that cannot be
 723 captured by a VLE table indexed on feed molality alone.

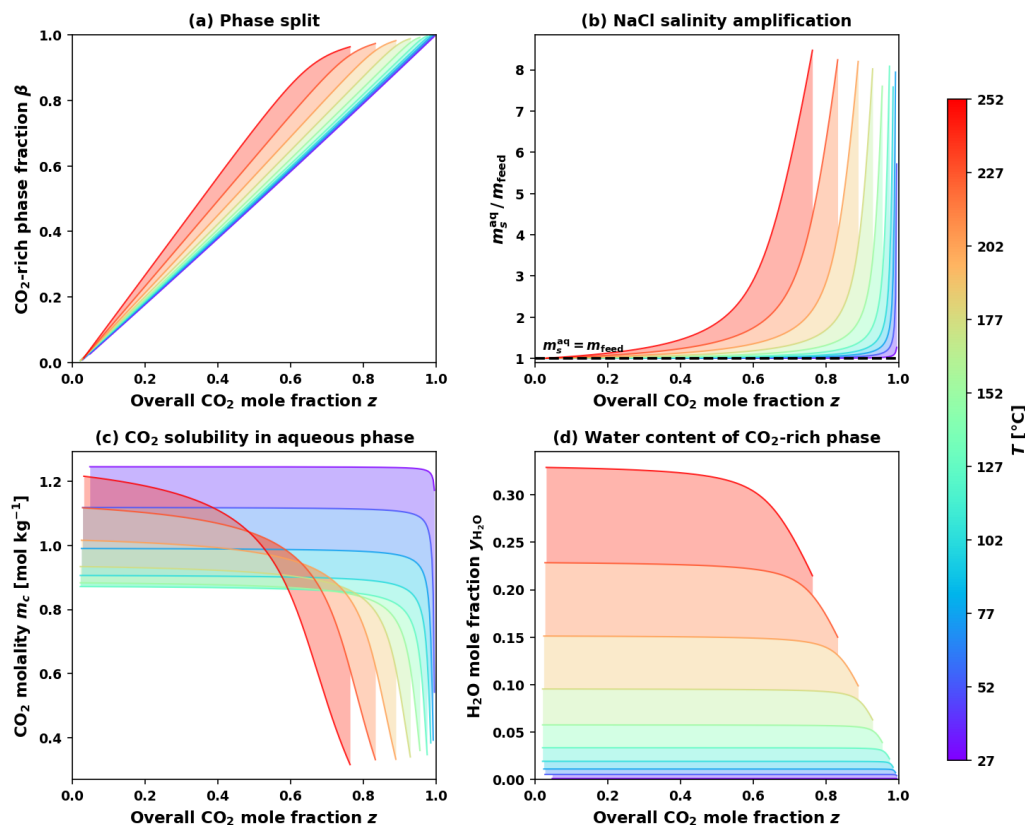


Figure 4: eCPA flash results as a function of overall CO₂ feed mole fraction z_{CO_2} at $P = 200$ bar, feed molality $m_s = 1.0$ mol kg⁻¹ NaCl, and ten isotherms from 27 to 252 °C (violet to red). (a) CO₂-rich phase fraction β ; the two-phase window narrows progressively at higher temperature. (b) Ratio of equilibrium aqueous NaCl molality to feed molality, m_s^{aq}/m_s ; NaCl concentrates in the aqueous phase as H₂O partitions to the CO₂-rich phase, reaching a factor of ~ 8 at $T = 252$ °C and $z_{CO_2} \approx 0.98$. (c) CO₂ molality m_{CO_2} in the aqueous phase; the salting-out effect driven by the elevated m_s^{aq} reduces CO₂ solubility with increasing z_{CO_2} . (d) H₂O mole fraction y_{H_2O} in the CO₂-rich phase; substantial water content at high temperatures (up to ~ 30 mol% at 252 °C) and its strong variation with z_{CO_2} highlight the need for full flash rather than a fixed-salinity VLE lookup.

724 Figure 4 demonstrates these effects for $P = 200$ bar, $m_s = 1.0$ mol kg⁻¹
 725 NaCl, and ten isotherms from 27 to 252 °C (in 25 °C increments) over the two-

726 phase feed range $z_{\text{CO}_2} = 0.005\text{--}0.995$. Panel (a) shows the CO_2 -rich phase
 727 fraction β , which rises from 0 to 1 across the two-phase window. Higher
 728 temperatures narrow the window because CO_2 is less soluble and H_2O more
 729 volatile, both effects shrinking the range of z_{CO_2} that sustains two-phase
 730 coexistence.

731 Panel (b) illustrates the salinity amplification: at the highest isotherm
 732 ($T = 252^\circ\text{C}$) the ratio m_s^{aq}/m_s reaches ≈ 8 at the high- z_{CO_2} end of the two-
 733 phase window, meaning the aqueous NaCl concentration increases eightfold
 734 relative to the injected brine as H_2O partitions into the growing CO_2 -rich
 735 phase. At lower temperatures the amplification is smaller because less H_2O
 736 leaves the aqueous phase at the lower β values attained.

737 Panel (c) shows the resulting CO_2 molality in the aqueous phase, m_{CO_2}
 738 $[\text{mol kg}^{-1}]$. Despite operating at fixed (T, P) , m_{CO_2} decreases substantially
 739 with increasing z_{CO_2} : at $T = 227^\circ\text{C}$ it falls from ≈ 1.1 to $\approx 0.4 \text{ mol kg}^{-1}$
 740 across the two-phase window. This is a direct consequence of the salting-out
 741 effect — the elevated m_s^{aq} suppresses CO_2 solubility. A simple VLE lookup
 742 at the feed molality m_s would return the wrong CO_2 solubility for all but the
 743 dilute- CO_2 end of the window.

744 Panel (d) shows the H_2O mole fraction in the CO_2 -rich phase, $y_{\text{H}_2\text{O}}$, and
 745 reveals a second consequence of the salting-out chain. As z_{CO_2} increases,
 746 β increases and m_s^{aq} rises (panel b), which suppresses CO_2 solubility in the
 747 aqueous phase (panel c): the aqueous phase can dissolve less CO_2 , so propor-
 748 tionally more CO_2 remains in the CO_2 -rich phase and, by molar conservation,
 749 $y_{\text{H}_2\text{O}}$ is diluted downward. At low temperatures this effect is invisible because
 750 the baseline water content of the CO_2 -rich phase is tiny; at high tempera-
 751 tures, where $y_{\text{H}_2\text{O}}$ starts near $\approx 35 \text{ mol}\%$, the same salting-out suppression
 752 translates into a large absolute drop — falling to $\approx 22 \text{ mol}\%$ by $z_{\text{CO}_2} \approx 0.75$
 753 at $T = 252^\circ\text{C}$. This also underscores the importance of accurate modeling
 754 of the CO_2 -rich phase at elevated temperatures.

755 Taken together, these results demonstrate that the correct representa-
 756 tion of CO_2 +brine phase behavior in reservoir models requires a full stabil-
 757 ity+flash computation for each cell at each time step, with the NaCl molality
 758 updated self-consistently from the aqueous-phase amount.

759 4.4. Aqueous-phase density

760 We validate pure-water density by comparing eCPA predictions (with the
 761 temperature-dependent P  neloux shift described in Section 2.3) against the
 762 IAPWS-95 international standard [42] — fitted to a comprehensive body

of experimental measurements and valid to 1 000 MPa and 1 273 K, with a documented uncertainty in liquid-water density of ± 0.001 – 0.02% . Over 475 liquid/compressed-water conditions spanning $T = 0$ – 350°C and $P = 1$ – $2\,000$ bar, the AARE is **0.30%**, with a maximum of 3% near 350°C (close to the critical point of water). For typical subsurface CO_2 storage conditions ($T \leq 120^\circ\text{C}$, $P = 75$ – 600 bar), the AARE is **0.12%**. We note that IAPWS-95 is itself a 56-parameter empirical Helmholtz free energy fit with no molecular basis, optimized specifically to reproduce water data. Achieving 0.30% with a physics-based association EoS (CPA/eCPA) is therefore a strong result. Full details, the parity plot, and the signed-error surface are provided in Section S4.1 of the Supplemental Information.

5. Computational Performance

5.1. Flash convergence statistics

5.1.1. Parameter-space scan for the salt-free CPA binary

To rigorously evaluate the robustness and efficiency of the accelerated SSI and multi-guess stability test, we perform a comprehensive parameter-space scan over $86 \times 18 \times 19 = 29,412$ conditions spanning $T = 0$ – 425°C , $P = 1$ – 1500 bar, and $z_{\text{CO}_2} = 0.001$ – 0.999 (Table 3). At each (T, P, z_{CO_2}) point, we run the stability test with all six initial guesses, followed by four independent flash strategies: standard SSI with Wilson K initialization (“Std+Wilson”), accelerated SSI with Wilson K (“Acc+Wilson”), accelerated SSI with K from the stability test (“Acc+StabK”), and the full hierarchical robust algorithm (“Robust”).

Table 3: Parameter-space scan grid for the salt-free CPA binary.

Variable	Range	Points	Spacing
T ($^\circ\text{C}$)	0–425	86	5°C
P (bar)	1–1500	18	semi-log
z_{CO_2}	0.001–0.999	19	variable
Total		29,412	

Of the 29,412 conditions, 19,586 (66.6%) are classified as single-phase by the stability test and 9,825 (33.4%) as two-phase. Table 4 compares six flash strategies on the two-phase subset. The hierarchical robust algorithm

789 achieves 100% convergence by combining the stability-informed K-value ini-
790 tialization with Wilson-K fallback for the 0.3% of points where the stability
791 K fails to converge. Only a single point at the extreme corner of the domain
792 ($T = 425^\circ\text{C}$, $P = 1500\text{ bar}$, $z_{\text{CO}_2} = 0.995$) produces a spurious instability;
793 all neighbors are correctly identified as single-phase.

794 The accelerated SSI reduces the mean iteration count from 25.4 to 13.7
795 (a 46% reduction, or $1.9\times$ speedup) compared to standard SSI with the same
796 Wilson K initialization (Figs. 6a and S11). The speedup is most pronounced
797 near the CO_2 critical point ($T \approx 31^\circ\text{C}$, $P \approx 74\text{ bar}$) and at extreme com-
798 positions, where standard SSI requires many iterations to converge (Fig. 5).
799 Elevated iteration counts at the lowest temperatures in the scan ($T \lesssim 12^\circ\text{C}$)
800 are also noteworthy: these conditions fall within the CO_2 hydrate stability
801 zone, where eCPA — which computes fluid-phase VLE only — returns a
802 metastable fluid equilibrium that may be physically unrealizable. The ad-
803 ditional iterations in this regime likely reflect the difficulty of converging a
804 fluid-phase flash at conditions where the true equilibrium involves a solid hy-
805 drate phase. Using K-values from the stability test as the flash initial guess
806 provides a further 8% reduction in iterations (13.7 to 12.6) on average.

Table 4: Comparison of six flash strategies across 9,825 two-phase conditions from the parameter-space scan ($T = 0\text{--}425^\circ\text{C}$, $P = 1\text{--}1500\text{ bar}$, $z_{\text{CO}_2} = 0.001\text{--}0.999$). The first four rows use cold-start initialization; the fifth and sixth use K-values interpolated from a pre-computed solution table extended to $T = 0\text{--}400^\circ\text{C}$ (99.9% of two-phase points). The sixth row additionally applies Newton polish once $\|\mathbf{g}\| < 10^{-4}$, with mean 4.8 SSI and 2.1 Newton iterations per converged call.

Strategy	Converged	Conv. rate	Mean iters	Median iters
Std SSI + Wilson K	9,756	99.3%	25.4	17
Acc SSI + Wilson K	9,775	99.5%	13.7	9
Acc SSI + Stability K	9,793	99.7%	12.6	9
Acc SSI + Stability K + Wilson K fall-back	9,825	100%	13.0	9
Acc SSI + Table K	9,818	99.9%	9.8	6
Acc SSI + Table K + Newton	9,808	99.8%	6.9	5

807 When a pre-computed solution table is available—as in sequential simu-
808 lation where flash solutions from the previous time step or neighboring grid

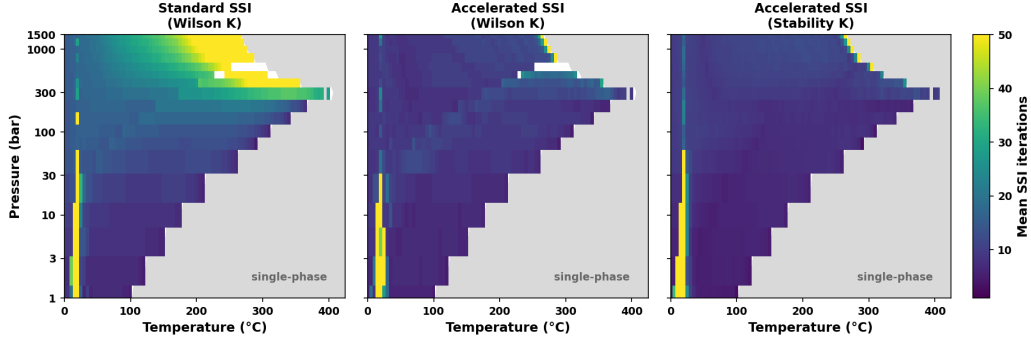


Figure 5: Mean SSI iteration count for the salt-free CPA flash, averaged over all z_{CO_2} values (two-phase points only), as a function of T and P . Left: standard SSI with Wilson K initialization. Center: accelerated SSI with Wilson K. Right: accelerated SSI with stability-test K. The accelerated SSI substantially reduces iteration counts across the entire parameter space, with the largest improvements near the CO_2 critical point ($T \approx 31^\circ\text{C}$, $P \approx 74\text{ bar}$). Elevated counts at the lowest temperatures ($T \lesssim 12^\circ\text{C}$) also reflect conditions within the CO_2 hydrate stability zone, where eCPA computes a metastable fluid-phase equilibrium.

cells serve as initial guesses—the iteration count drops further to a mean of 9.8 (median 6), a 25% improvement over the best cold-start strategy. The table covers $T = 0\text{--}360^\circ\text{C}$ with 1°C resolution and all practically relevant pressures ($P = 1\text{--}2000\text{ bar}$); it captures 99.9% of the two-phase scan points, eliminating the Wilson-K fallback for virtually all conditions. Because the table also pre-classifies each cell as single- or two-phase, the Michelsen stability test (six guesses, $\approx 10\text{ ms}$ per guess) can be skipped for all cells confidently inside the two-phase window, so the dominant cost of a cold-start stability+flash call ($\approx 30\times$ higher wall time) is avoided in the warm-start regime. Phase compositions obtained from the two approaches agree to within 1.1% AARE for x_{CO_2} and 3.9% for $y_{\text{H}_2\text{O}}$, with residual discrepancies concentrated near the two-phase boundary where both methods are most sensitive to initial guesses.

Adding Newton polish (Table 4, bottom row) reduces the mean count further to 6.9 (4.8 SSI + 2.1 Newton), a 29% improvement over table-warm-started SSI alone and a 73% reduction from the standard cold-start. Newton polish is triggered at 85.6% of points; the remaining 14.4% converge in fewer than $\tau_{\text{switch}} = 10^{-4}$ SSI steps and need no Newton correction. In the near- CO_2 -critical region ($T \approx 7\text{--}57^\circ\text{C}$) the Newton step can temporarily diverge; the solver then reverts to the pre-Newton SSI state and continues, so no

convergence is lost in that region. The iteration breakdown and wall-time ratio across the full (T, P) parameter space are shown in Fig. S10.

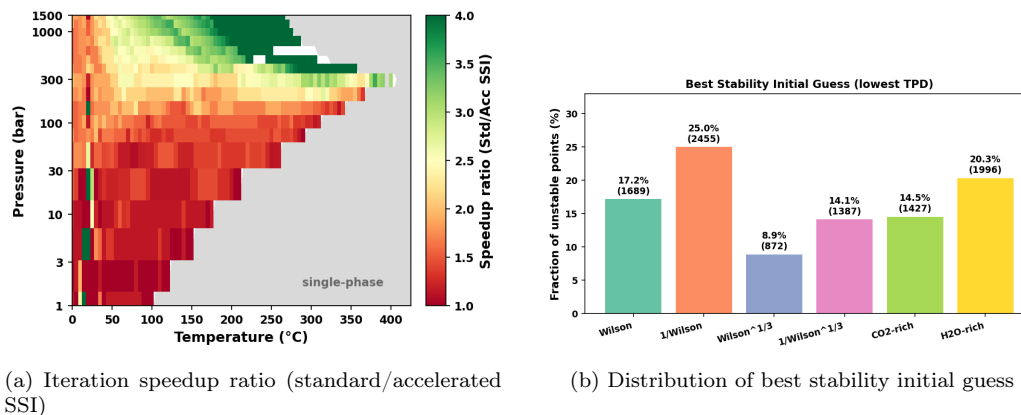


Figure 6: (a) Iteration speedup from accelerated SSI, averaged over z_{CO_2} (two-phase only). The speedup exceeds $2\times$ across most of the parameter space and reaches $3\text{--}4\times$ near the critical region. (b) Distribution of the initial guess that found the lowest TPD across all 9,825 unstable points. No single guess dominates: each of the six trials is the best for 9–25% of conditions, demonstrating the importance of multiple initial guesses for robust stability testing.

Figure 6b shows that all six stability initial guesses contribute materially: Wilson is best for 17.2% of unstable points, inverse Wilson for 25.0%, H₂O-rich for 20.3%, CO₂-rich for 14.5%, inverse cube-root for 14.1%, and cube-root Wilson for 8.9%. While the first two guesses (Wilson and inverse Wilson) can detect instability ($\text{TPD} < 0$) for 99.8% of conditions, they locate the *most negative* TPD—and thus the best K-value initialization for the subsequent flash—for only 42.2%. All six guesses are required to ensure the optimal K-value initialization across the full parameter space.

5.1.2. Convergence statistics for the eCPA ternary flash

To characterize eCPA flash performance over the full thermodynamic parameter space relevant to geological CO₂ storage and geothermal energy production, we compute the three-dimensional solution table described in Section 3.6.1, covering the grid summarized in Table 5.

Each grid point is evaluated with the full hierarchical stability+flash algorithm: the Michelsen stability test (six trial compositions) is performed first; only genuinely unstable conditions proceed to the K-value SSI flash.

847 The CO_2 feed fraction z_{CO_2} is not a table dimension; a representative value
848 $z_{\text{CO}_2} = 0.5$ is used for the stability call (K-values and tie-line endpoints are
849 only weakly z_{CO_2} -dependent, as discussed in Section 3.6.1). Of the 505,400
850 grid points, 36,100 use the CPA sub-model ($m_s = 0$, no electrostatic terms);
851 the eCPA model covers the remaining 469,300 points.

Table 5: Parameter grid for the eCPA ternary solution table ($\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$, $361 \times 100 \times 14 = 505,400$ points).

Dimension	Range	Points
Temperature T	0–360 °C	361 (1 °C steps)
Pressure P	1–2000 bar	100 (log-spaced)
NaCl feed molality m_s	0–6 mol kg ⁻¹	14

852 Table 6 reports the outcome of all evaluations. Every grid point is
853 classified—100% convergence is achieved over the complete three-dimensional
854 domain.

Table 6: Classification of eCPA flash outcomes over the full 3D solution table (505,400 total points; 469,300 eCPA, 36,100 CPA).

Classification	Count	Description
Two-phase, converged	289,065	Valid two-phase split; SSI converged
Single-phase (stable)	216,335	Michelsen stability test: no phase split
Total classified	505,400	100%

855 *CO₂ solubility in the aqueous phase.* Figures 7 and 8 map x_{CO_2} and $y_{\text{H}_2\text{O}}$,
856 respectively, across the full (T, P) domain for six NaCl molalities (and a pure
857 salt-free condition with CPA) at a representative feed $z_{\text{CO}_2} = 0.5$; red curves
858 trace the dew- (solid) and bubble-point (dashed) boundaries, with the gray
859 region indicating single-phase conditions. Figure 9 maps mean flash wall time
860 per call across the T – P plane for all six NaCl molalities, serving as a proxy for
861 solver effort. Figure 7 shows x_{CO_2} (aqueous phase) over a logarithmic color
862 scale spanning 0.05–25 mol%. The two-phase window is broad, spanning the
863 full pressure range from 1 to 2000 bar at most temperatures below ~ 250 °C,
864 and closes at the highest temperatures as CO_2 and water approach complete

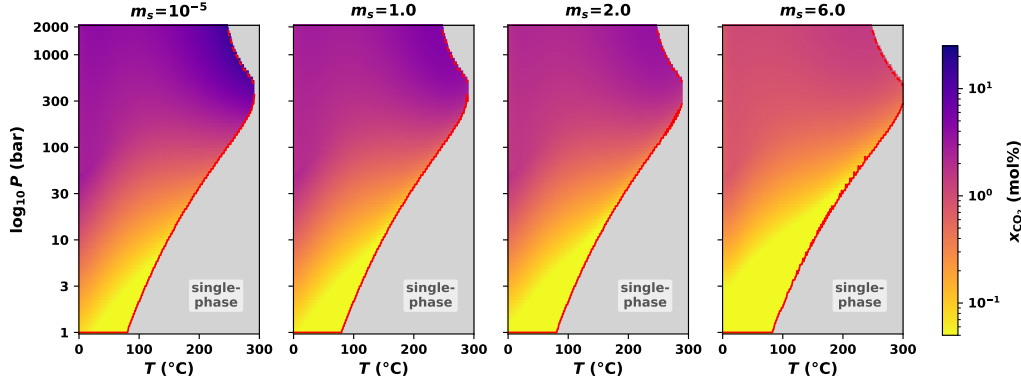


Figure 7: Equilibrium CO_2 mol% in the aqueous phase (x_{CO_2}) over the T – P plane at representative feed $z_{\text{CO}_2} = 0.5$, for six NaCl molalities (columns, $m_s = 0$ – 6 mol kg^{-1}). The logarithmic color scale spans three orders of magnitude to resolve both the dilute near-bubble-point regime and the high-solubility hydrothermal limit. Grey regions are thermodynamically single-phase at those conditions. Red curves mark the phase envelope: the solid line is the dew-point boundary (lower pressure limit of the two-phase window) and the dashed line is the bubble-point boundary (upper pressure limit). The phase boundaries are nearly independent of m_s ; the salting-out effect manifests as a reduction in x_{CO_2} within the window, not as a shift of its extent.

865 miscibility. Three physically distinct solubility regimes are apparent. At low
866 temperatures ($T \lesssim 50^\circ\text{C}$), the CO_2 -rich phase is a dense supercritical or
867 liquid-like fluid; solubility is moderate and increases with pressure. At inter-
868 mediate temperatures (50 – 200°C), solubility passes through a minimum as
869 the competing effects of pressure and thermal expansion partially cancel. At
870 high temperatures ($T \gtrsim 250^\circ\text{C}$), solubility rises sharply because the aqueous
871 phase increasingly resembles steam; values reach 20 – $25 \text{ mol}\%$ at the high-
872 est pressures, well into a regime relevant to hydrothermal and geothermal
873 CO_2 storage. Increasing NaCl molality (left to right) reduces CO_2 solubil-
874 ity within the two-phase window—the classical salting-out effect. The phase
875 boundaries (red curves) are nearly independent of m_s : the dew-point shifts
876 by less than 15% across the full salinity range, and the bubble-point (visible
877 only near $T \approx 250^\circ\text{C}$ where it falls below the 2000 bar grid ceiling) likewise
878 shows negligible salinity dependence. The dominant effect of NaCl is there-
879 fore a reduction in x_{CO_2} within the two-phase window rather than a change
880 in the window’s extent.

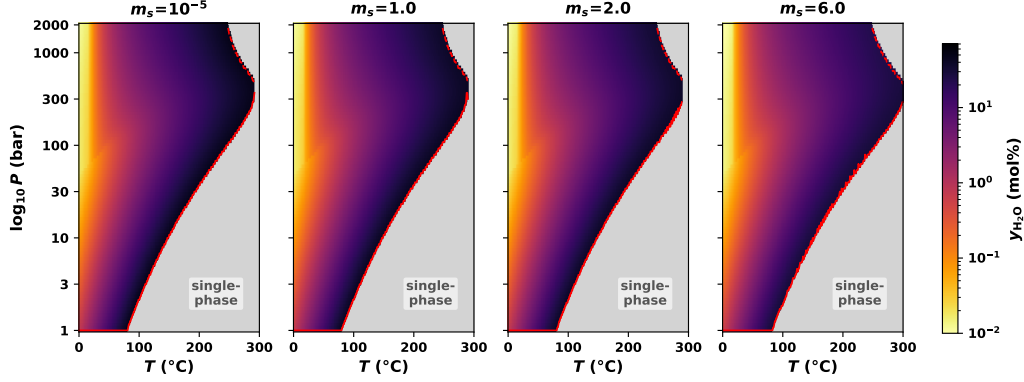


Figure 8: Equilibrium H_2O mol% in the CO_2 -rich phase ($y_{\text{H}_2\text{O}}$) over the T - P plane at representative feed $z_{\text{CO}_2} = 0.5$, on the same grid as Fig. 7. Unlike x_{CO_2} , water content in the CO_2 -rich phase is governed almost entirely by temperature (water vapour pressure), producing near-vertical isopleths that are nearly independent of pressure and NaCl molality. Red curves show the phase envelope (solid: dew point; dashed: bubble point); their near-identical positions across columns confirm that NaCl has negligible influence on the phase boundaries. At temperatures above 250°C the CO_2 -rich phase contains more than 25 mol% H_2O , approaching 70 mol% at the highest temperatures—a regime relevant to geothermal CO_2 storage.

881 *H_2O content of the CO_2 -rich phase.* Figure 8 shows $y_{\text{H}_2\text{O}}$ (CO_2 -rich phase)
 882 on the same grid layout. In sharp contrast to CO_2 solubility, the water con-
 883 tent of the CO_2 -rich phase is controlled almost entirely by temperature: the
 884 color field in each panel varies primarily along the horizontal axis, with near-
 885 vertical isopleths. This reflects the dominant role of the water saturation
 886 pressure, which rises steeply with temperature and sets the water activity
 887 in the CO_2 -rich phase regardless of pressure or salinity. At $T = 50^\circ\text{C}$, $y_{\text{H}_2\text{O}}$
 888 is below 1 mol% across essentially all pressures; by $T = 300^\circ\text{C}$ it reaches
 889 50–70 mol%, meaning the nominal CO_2 -rich phase is in fact predominantly
 890 water vapour. The effect of NaCl (columns) on $y_{\text{H}_2\text{O}}$ is weak: the isopleths
 891 and the phase envelope (red curves) are essentially unchanged across all salin-
 892 ity columns, confirming that water content in the CO_2 -rich phase is set by
 893 temperature alone and is insensitive to brine salinity.

894 *Flash wall time.* Figure 9 maps the mean wall time per eCPA flash call
 895 (table-warm-started K-value SSI) across the T - P plane for six NaCl molal-
 896 ities; gray regions are single-phase. The $m_s = 0$ (CPA) column is excluded:
 897 its cold-start six-trial robust solver runs at a median of ~ 215 ms per call

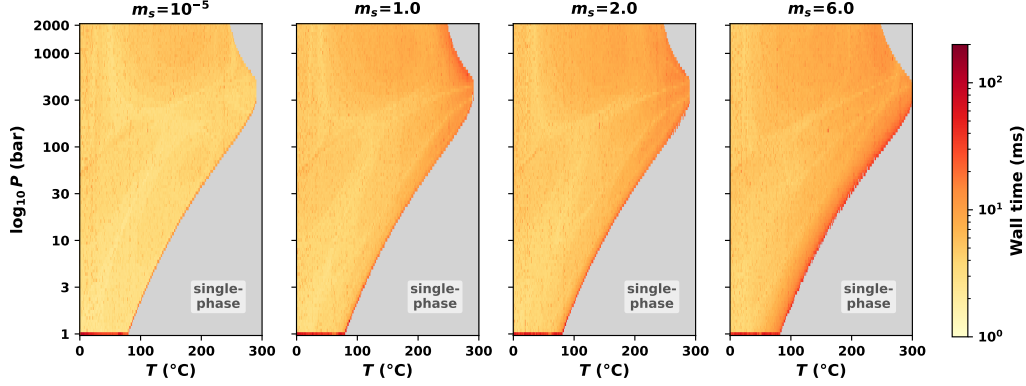


Figure 9: Mean wall time per flash call (ms) for the eCPA warm-started K-value SSI over the T - P plane for six NaCl molalities (columns), from the 3D scan. Gray regions are thermodynamically single-phase. The colormap is log-scaled (1–200 ms). Times are lowest in the interior of the two-phase window and highest near the phase boundaries where K-values approach unity and more SSI iterations are required. The $m_s=0$ (CPA) column is excluded; its cold-start robust solver runs at a median of ~ 215 ms per call and is reported separately in Table 7.

(reported in Table 7), which would render uniformly at the colorbar ceiling; the warm-start table applies equally to CPA. Wall time serves as a proxy for solver effort. Times are lowest ($\lesssim 5$ ms) in the interior of the two-phase region, where the warm-started SSI converges in 2–3 steps. Times rise near the phase boundaries—particularly near the bubble point and the CO_2 critical temperature—where K-values approach unity and more iterations are required. The pattern is qualitatively similar across all NaCl molalities.

Table 7 summarizes SSI iteration counts and wall times across the experimental validation datasets used in Sections 4.1–4.2.

Table 7: Flash convergence statistics across the experimental validation datasets.

Method	N calls	Mean iter	Median	Max	Median time/call
CPA (cold, accel.)	29,412	6.3	6	272	~ 215 ms
CPA (warm SSI)	626	6.2	6	40	~ 2 ms
eCPA (warm SSI, $m_s \approx 0$)	631	5.9	6	10	~ 2 ms
eCPA (warm SSI)	686	5.8	6	16	~ 2 ms

907 All three warm-start rows use the same solution table and achieve a me-
 908 dian of 6 SSI iterations and $\lesssim 2$ ms per call. The warm-started CPA row
 909 directly demonstrates that the $\sim 100\times$ speedup over cold-start CPA (row 1)
 910 is attributable entirely to the warm-start strategy: the eCPA model adds
 911 electrostatic terms not present in CPA, yet both achieve comparable wall
 912 times when started from a good initial guess. Among the eCPA rows, the
 913 additional salt does not materially increase cost. Each warm-started SSI step
 914 costs only ~ 0.2 ms; the solution-table lookup itself takes $\lesssim 0.05$ ms and con-
 915 tributes negligible overhead. Convergence is robust across the full validation
 916 range.

917 5.2. Inner EoS solve efficiency: Newton–Raphson versus cold-start alterna- 918 tives

919 Each outer SSI iteration requires solving the inner per-phase EoS system.
 920 The two systems — CPA’s $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ and eCPA’s $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ — dif-
 921 fer in structure but share the same warm-started Newton–Raphson strategy
 922 described in Sections 3.3 and 3.4; the analytical Jacobians are derived in Sec-
 923 tions S3.1 and S3.2 of the Supplemental Information. The benchmarks below
 924 quantify the advantage of the warm-started Newton solve over the respective
 925 cold-start alternatives, and are distinct from the outer SSI+Newton-polish
 926 acceleration discussed in Section 5.1 and Fig. S10.

927 *CPA: $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ system.* Starting from the previous SSI iterate (typi-
 928 cally within 0.5% of the converged value), the warm-started Newton solve
 929 converges in **3–4 iterations** (median 3, maximum 4 across 9,825 two-phase
 930 conditions). The cold-start fallback — a 12-point log-spaced scan to bracket
 931 the root of F_1 , followed by Brent refinement — requires on average 23 resid-
 932 ual evaluations (median 21) per call, equivalent to ≈ 0.38 ms. The warm-
 933 start Newton solve costs ≈ 0.03 ms across 128 two-phase conditions spanning
 934 $T = 27\text{--}227^\circ\text{C}$ and $P = 50\text{--}800$ bar: a **13** $\times$ reduction in inner-solve cost per
 935 SSI step.

936 *eCPA: $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ system.* For the aqueous 3×3 system, a 0.5–1% warm-
 937 start perturbation yields convergence in **4–6 iterations** (median 5 at 1%),
 938 with a wall time of ≈ 0.25 ms per call. The cold-start alternative — solving
 939 from a heuristic initial guess via a general-purpose modified-Powell root-
 940 finder — requires on average 47 residual evaluations and costs ≈ 1.5 ms per
 941 call: a **6** $\times$ overhead.

942 *Computational cost of CPA and eCPA versus a cubic EOS.* To quantify the
 943 per-iteration overhead of the association terms relative to a plain cubic EoS,
 944 we time each component of a single SSI iteration across 871 two-phase (T, P)
 945 conditions sampled from the CO₂–H₂O parameter space. CPA (with the
 946 Newton Z/χ solve) is compared against a notional SRK reference (same
 947 EoS parameter computation and fugacity evaluation, but Z found by direct
 948 solution of the cubic and $\chi \equiv 1$). The per-iteration median wall times were
 949 98 μ s (CPA) versus 32 μ s (SRK), a ratio of **3.1** $\times$. The EoS-parameter and
 950 fugacity-evaluation steps are identical between the two models; the entire
 951 overhead arises in the Z/χ solve, which costs 75 μ s (CPA Newton, two phases)
 952 versus 9 μ s (SRK direct cubic, two phases). This factor-of-3 per-iteration cost
 953 must be weighed against the additional physics captured by CPA: accurate
 954 partitioning of a self- and cross-associating system over a wide temperature
 955 and pressure range, and (in the full eCPA model) correct salt activity without
 956 empirical brine corrections.

957 5.3. Simplified versus full flash: accuracy and efficiency

958 Over 3,285 two-phase conditions spanning $T = 10$ – 250 °C, $P = 25$ –
 959 800 bar, and $m_s = 1$ – 6 mol kg^{−1}, the simplified flash reproduces the full K-
 960 value SSI result with AARE < 2% in dissolved CO₂ molality for $T \leq 80$ °C,
 961 rising to 3.2% at 100 °C and exceeding 5% above 120 °C as the $y_{\text{H}_2\text{O}} = 0$
 962 assumption deteriorates. Phase identification agrees with the full flash in
 963 99.7% of cases. The accuracy is insensitive to brine salinity: the error is con-
 964 trolled entirely by temperature. Contrary to expectation, the mean wall time
 965 is essentially identical to the full flash (0.8 $\times$ speedup), because Brent root-
 966 finding on the 1D fugacity residual requires ~ 11 inner Newton evaluations
 967 while SSI needs only ~ 6 outer iterations with at most two inner solves each.
 968 Full per-temperature statistics, diagnostic figures, and a salinity breakdown
 969 are provided in Section S4.2 of the Supplemental Information.

970 6. Reservoir Simulation Demonstration

971 To confirm that the flash algorithms developed in this work perform re-
 972 liably under realistic conditions, we implement them within an isothermal
 973 IMPEC compositional simulator for CO₂ injection into heterogeneous 3D
 974 saline aquifers. Across 300 time steps and all 2500 grid cells of a 50×50
 975 domain, no flash failures are recorded in either the salt-free (CPA, $m_s = 0$)
 976 or saline (eCPA, $m_s = 4$ mol kg^{−1}) case, confirming 100% convergence of

the hierarchical stability and flash approach. The warm-started eCPA flash achieves a throughput of $14,270 \text{ calls s}^{-1}$ and the CPA flash $8,840 \text{ calls s}^{-1}$, with the flash accounting for 80–87% of total step cost. Full details of the numerical scheme, simulation setup, spatial results, and performance metrics are provided in Section S6 of the Supplemental Information.

7. Conclusions

This work presents the first complete phase stability and flash framework built on the eCPA equation of state of Coelho et al. [9], extending it from single-phase property evaluation to fully converged, production-ready VLE calculations for $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ mixtures across reservoir to hydrothermal conditions.

The central algorithmic contribution is the tight integration of four independently novel elements:

- Fast inner Newton–Raphson solvers for the CPA and eCPA root-finding problems, based on analytical Jacobians derived in closed form, which reduce the per-evaluation overhead to only $\sim 3\times$ that of a simple cubic EoS.
- A hierarchical outer flash algorithm combining Michelsen stability testing with six initial guesses, accelerated SSI, and optional Newton polish — achieving 100% convergence over a comprehensive parameter-space scan.
- A precomputed three-dimensional (T, P, m_s) solution table that provides warm starts reducing flash iterations by a factor of three relative to cold-start SSI.
- A simplified flash that exploits the $y_{\text{H}_2\text{O}} \approx 0$ approximation to reduce the equilibrium problem to a single scalar equation, yielding errors below 2% in dissolved CO_2 for $T \leq 80^\circ\text{C}$ at essentially the same computational cost.

Together, these enable warm-started eCPA flash at throughputs competitive with conventional cubic-EoS implementations, as demonstrated in a prototype IMPEC reservoir simulator.

Validation against >1100 experimental data points shows AARE of 6.5% for CO_2 solubility under subsurface conditions and 6.6% for CO_2 molality in

1010 NaCl brine. Comparison against the IAPWS-95 international standard over
1011 475 liquid conditions ($T = 0\text{--}350\text{ }^{\circ}\text{C}$, $P = 1\text{--}2000\text{ bar}$) with a temperature-
1012 dependent P  neloux volume shift for H_2O gives a density AARE of 0.30%
1013 overall and 0.12% for subsurface storage conditions.

1014 **Future Work**

1015 A fundamental extension is the generalization to brines containing mul-
1016 tiple salts (e.g. $\text{NaCl} + \text{CaCl}_2 + \text{MgCl}_2$, as encountered in deep formation
1017 waters). In the single-salt eCPA framework the NaCl molality provides a
1018 single scalar coupling between phase amounts and phase compositions. For
1019 multi-salt brines the ionic interactions become more complex, and it will
1020 likely be necessary to shift from a molecular-species basis to an ionic basis,
1021 treating each cation and anion as an independent component with its own
1022 association and electrostatic parameters. Extending the eCPA framework
1023 in this direction represents a significant but tractable step toward a truly
1024 general thermodynamic engine for saline CO_2 storage.

1025 The algorithms and validation presented here are implemented entirely in
1026 Python. While this facilitates rapid prototyping and transparent benchmark-
1027 ing, it means that the absolute throughput numbers reported are conservative
1028 relative to what a compiled implementation would achieve. A natural next
1029 step is to integrate the CPA and eCPA flash routines into a production-level
1030 compiled simulator that fully accounts for gravity-driven and density-driven
1031 flow, Fickian diffusion, capillarity, and explicitly resolved discrete fractures
1032 in three-dimensional unstructured grids (e.g. Moortgat et al. 43, 20, Moort-
1033 gat and Firoozabadi 44, 45, Moortgat 17, Solovsk   and Firoozabadi 46, 3).
1034 Such an integration will enable high-fidelity simulation of CO_2 sequestration
1035 in saline aquifers at reservoir scale.

1036 **Acknowledgements**

1037 Abbas Firoozabadi was supported by the member companies of the Reser-
1038 voir Engineering Research Institute. J. Moortgat did not receive any specific
1039 grant from funding agencies in the public, commercial, or not-for-profit sec-
1040 tors.

Author Contributions

Joachim Moortgat: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft. **Felipe Mourão Coelho:** Data curation, Writing – original draft. **Abbas Firoozabadi:** Conceptualization, Funding acquisition, Writing – original draft.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with writing, editing, and LaTeX formatting of the manuscript. After using this tool, the authors reviewed and edited all content as needed and take full responsibility for the content of the published article.

Data and Code Availability

All source code implementing the CPA and eCPA equations of state, the phase stability and flash algorithms, the solution table construction, and the reservoir simulator described in this work is freely available at <https://github.com/buckai-observatory/ecpa-flash> (*upon acceptance of paper). The repository includes the precomputed three-dimensional solution table, all experimental VLE data used for validation, and scripts to reproduce every figure and table in this paper.

Supplementary Information

The following material is provided as Supplementary Information.

- **Section S1 (Nomenclature):** Complete listing of all mathematical symbols used throughout the paper, organized by type (Latin, Greek, subscripts, and superscripts).
- **Section S2 (EoS Parameters):** Tables of pure-component and binary interaction parameters for the CPA/eCPA model (Tables S2–S3), and coefficients for the temperature-dependent Péneloux volume shift for H₂O (Table S4; Eq. S1).

- 1070 • **Section S3 (Analytical Jacobians):** Closed-form Jacobian matrices for the two inner Newton–Raphson systems: the 3×3 CPA system in $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ and the 3×3 eCPA aqueous-phase system in $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ (Eqs. S2–S29).
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- 1074 • **Section S4 (Additional Validation Figures):** $\text{CO}_2+\text{H}_2\text{O}$ phase equilibria at all 35 temperatures not shown in the main text (Fig. S1); CO_2+NaCl phase equilibria at 18 temperatures (Figs. S2–S3); AARE heatmaps for CO_2+NaCl (Fig. S4) and $\text{CO}_2+\text{H}_2\text{O}$ (Fig. S7); extended validation of aqueous density against the IAPWS-95 standard (Figs. S5–S6); and accuracy and efficiency assessment of the simplified flash (Table S5; Figs. S8–S9).
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- 1081 • **Section S5 (Computational Performance Figures):** Supplementary figures on SSI+Newton iteration breakdown (Fig. S10) and flash strategy comparison (Fig. S11), supplementing the convergence and efficiency analysis of Section 5.
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- 1085 • **Section S6 (Reservoir Simulation Details):** Full description of the numerical scheme and simulation parameters (Table S6; Figs. S12–S13; Table S7), supplementing Section 6.
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